MATHEMATICS GRADE 10: LINEAR MOTION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.4 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 3.0: Measurements and Geometry
Sub-Strand	Sub-Strand 3.4: Linear Motion
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Distance and displacement: definitions, units, and real-life examples (e.g., running from Nairobi CBD to Uhuru Park) – Speed and velocity: definitions, formulae, units, and distinction between scalar and vector quantities – Acceleration and deceleration: definitions, formulae, and real-life contexts (e.g., a matatu accelerating out of a bus stop) – Calculations involving velocity and acceleration in different situations – Displacement–time graphs: drawing, reading, and interpreting (stationary, uniform motion, non-uniform motion) – Velocity–time graphs: drawing from given data tables, reading, and interpreting (uniform velocity, uniform acceleration, deceleration) – Area under a velocity–time graph as a measure of displacement – Gradient of displacement–time and velocity–time graphs and their physical meaning – Relative speed of two bodies moving in the same direction and in opposite directions – Application of linear motion concepts to real-life situations (athletics, road safety, Kenyan transport contexts)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) distinguish between distance and displacement, and between speed and velocity, using real-life Kenyan contexts; b) calculate speed, velocity, and acceleration for objects in linear motion and interpret the results; c) draw accurate displacement–time graphs and velocity–time graphs from given data in different situations; d) interpret displacement–time graphs and velocity–time graphs to describe the motion of an object, including determining displacement from the area under a velocity–time graph; e) determine the relative speed of two bodies moving in the same or opposite directions and apply this knowledge to real-life situations such as road safety and athletics;
Core Competencies	– Communication and Collaboration: Learners discuss and debate the difference between distance/displacement and speed/velocity in pairs and groups, articulating reasoning clearly using mathematical language and presenting graph interpretations to peers. – Critical Thinking and Problem Solving: Learners analyse displacement–time and velocity–time graphs to deduce the nature of motion, calculate unknown quantities, and evaluate real-world scenarios such as road-safety stopping distances on Kenyan highways. – Digital Literacy: Learners engage with video resources on distance & displacement and simulations of motion graphs, developing the ability to extract quantitative information from digital media.

	– Self-Efficacy: Learners build confidence by moving from concrete timer-race observations to abstract graph interpretation, tracking their own growing understanding on the Driving Question Board. – Creativity and Imagination: Learners design their own motion scenarios (e.g., a boda-boda trip from Kisumu town to Dunga Beach) and represent them as displacement–time or velocity–time graphs, demonstrating creative application of concepts.
Core Values	– Responsibility: Learners connect the concept of speed limits and reaction distances to responsible road use on Kenyan roads, appreciating the life-saving importance of mathematics in everyday decisions. – Respect: Learners respect diverse contributions during group sensemaking discussions and honour the experiences of classmates who share observations from the timer-race activity. – Unity: Collaborative group work during graph-drawing and relative-speed investigations fosters a sense of shared purpose and teamwork across different learner abilities. – Social Justice: Learners examine how poor road infrastructure and unsafe vehicle speeds disproportionately affect vulnerable Kenyans, linking mathematical literacy to advocacy for safer communities. – Integrity: Learners record experimental timer-race data honestly and accurately, understanding that falsified data leads to incorrect conclusions — a foundation of mathematical and scientific integrity.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions during the timer-race phenomenon about why two runners covering the same distance can have different displacements, anchoring the unit's inquiry. – Planning and Carrying Out Investigations: Learners design and conduct the timer-race activity, measuring time and distance carefully and recording data in tables for subsequent graph construction. – Analysing and Interpreting Data: Learners analyse tabulated motion data to draw displacement–time and velocity–time graphs and interpret gradients and areas to extract physical meaning. – Using Mathematics and Computational Thinking: Learners apply formulae for speed, velocity, and acceleration; calculate gradients of graphs; find areas under velocity–time graphs; and solve relative-speed problems systematically. – Constructing Explanations: Learners use graph evidence and calculations to explain the motion of objects in real-life Kenyan contexts such as athletes at a school sports day or matatus on Thika Superhighway. – Obtaining, Evaluating, and Communicating Information: Learners read and critically evaluate written passages and video content on relative speed, then communicate findings through annotated graphs and written explanations.
Pertinent & Contemporary Issues (PCIs)	– Safety: Road safety is woven throughout the unit — learners calculate stopping distances and relative speeds to understand why speeding is dangerous on Kenyan roads, supporting the National Road Safety Programme. – Life Skills: Decision-making and problem-solving skills are developed as learners apply linear motion concepts to real scenarios such as planning travel time from Mombasa to Malindi. – Environmental Education: Learners discuss how vehicle speed and acceleration relate to fuel consumption and carbon emissions, linking mathematics to environmental stewardship in the Kenyan context. – Health Education: Physical activity during the timer-race phenomenon promotes learner health and wellness, and links are made to the biomechanics of Kenyan long-distance runners such as Eliud Kipchoge. – Financial Literacy: Learners explore how speed and distance calculations affect transport costs and logistics for Kenyan farmers moving produce from Rift Valley farms to Nairobi markets.
Career Connections	– Engineering (Civil/Transport): Understanding linear motion is fundamental to designing roads, railways (e.g., the SGR Nairobi–Mombasa line), and bridges with appropriate speed and safety margins. – Physics and Mathematics Teaching: Mastery of motion graphs and kinematics concepts prepares learners to teach these topics at secondary and tertiary levels across Kenya. – Athletics Coaching and Sports Science: Kenyan athletics coaches use speed, velocity, and acceleration data to optimise training programmes for world-class runners like Faith Kipyegon and Eliud Kipchoge.

	– Aviation and Navigation: Pilots and maritime navigators (e.g., on Lake Victoria) rely on displacement, velocity, and relative-speed calculations for safe and efficient travel. – Urban Planning and Traffic Management: Nairobi city planners and traffic engineers use motion data to design efficient road networks, manage congestion, and set speed limits. – Logistics and Supply Chain Management: Truck fleet managers and logistics companies in Kenya calculate travel times, speeds, and fuel requirements using linear motion mathematics.
Focus for Lessons	This unit anchors the study of linear motion in a vivid, kinaesthetic phenomenon — a timed race activity in the school compound — that immediately surfaces the puzzling distinction between how far learners travel and where they end up (distance vs. displacement). From this shared experience, learners progressively build a connected understanding of speed, velocity, and acceleration through data collection, graph construction, and graph interpretation, culminating in the powerful concept of relative speed applied to everyday Kenyan transport scenarios. The approach is inquiry-driven and context-rich, using Kenyan athletes, matatus, SGR trains, and boda-bodas to ensure every mathematical concept is grounded in a world learners recognise and care about.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter? KICD KEY INQUIRY QUESTIONS: 1. How do distance, displacement, speed, velocity, and acceleration differ from one another, and how are they calculated? 2. How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Timer Race Puzzle: At the start of the unit, learners participate in a 'timer race' in the school compound. Two groups of learners are asked to travel from the same starting point to the same finishing point in the same amount of time — but one group must walk in a straight line while the other must follow a winding zigzag path around obstacles. Both groups arrive at the finish at roughly the same time, yet learners quickly notice something puzzling: a third group member who walked the zigzag path and then retraced their steps back to a midpoint is also declared to have covered the 'same distance' as the straight-line walker, yet ends up at a completely different location. This creates genuine cognitive conflict: if distance is the same and time is the same, why are the runners in such different places? Why does the speedometer reading of a matatu not tell you which direction Nairobi is? Why can Eliud Kipchoge run 42 km in a marathon yet end up only a few hundred metres from where he started? Students observe that the same numerical value — whether for distance, speed, or time — can describe very different physical situations depending on direction. This puzzling observation motivates the entire unit: learners need a richer mathematical language (displacement, velocity, acceleration, and graphical representation) to fully describe and predict motion in the real world.
Storyline Thread	Lesson 1 — Anchoring Phenomenon (Timer Race): Learners conduct the timer race activity in the school compound, collecting time and distance data. They confront the puzzle of distance vs. displacement firsthand and post their first 'wonder questions' on the Driving Question Board (DQB). Lesson 2 — Distance, Displacement, Speed & Velocity (Video + Discussion): Learners watch a short video on distance & displacement, then a second clip on speed vs. velocity. They reconcile the video evidence with their timer-race experience, define

	key terms, and solve introductory calculations using Kenyan contexts (e.g., a cyclist riding from Kericho to Nakuru). Lesson 3 — Velocity, Speed & Acceleration (Illustrations + Calculations): Using illustrated scenarios of a matatu accelerating from a bus stop and a motorcycle braking, learners derive the formula for acceleration, distinguish positive from negative acceleration (deceleration), and practise multi-step calculations. The DQB is updated with new answers and new questions. Lesson 4 — Velocity and Acceleration Problems (Practice & Sensemaking): Learners work in groups on structured problem sets involving velocity and acceleration in varied situations — athletes, vehicles on Kenyan roads, and falling objects. Groups present solutions and justify their reasoning to the class. Lesson 5 — Drawing Displacement–Time Graphs: Using data tables generated from the timer-race activity and from the SGR supporting phenomenon, learners practise drawing accurate displacement–time graphs on grid paper. They identify the features of stationary, uniform, and non-uniform motion. Lesson 6 — Interpreting Displacement–Time Graphs: Learners receive a set of pre-drawn displacement–time graphs representing different journeys (a boda-boda in Kisumu, a jogger in Nairobi's Karura Forest) and interpret each graph — reading values, calculating gradient/speed, and writing narrative descriptions of each journey. Lesson 7 — Drawing Velocity–Time Graphs: Learners are given tabulated velocity–time data (including from Kipchoge's marathon splits) and practise drawing velocity–time graphs. They discover that the gradient represents acceleration and begin to calculate it graphically. Lesson 8 — Interpreting Velocity–Time Graphs & Area Under the Graph: Learners interpret a series of velocity–time graphs, calculate acceleration from gradients, and discover — through a guided investigation — that the area under the graph equals displacement. They apply this to determine stopping distances in the boda-boda braking scenario. Lesson 9 — Relative Speed (Reading + Illustration): Learners read a passage on relative speed and study illustrated scenarios of two matatus moving in the same direction and in opposite directions on a Kenyan highway. They derive the rules for combining velocities and solve relative-speed problems, updating the DQB with newly answered questions. Lesson 10 — Application & Cumulative Model Building: Learners tackle a rich real-life problem integrating all sub-strand concepts (e.g., planning the safe overtaking of a truck on the Nakuru–Nairobi highway). They finalise and present their cumulative motion models — annotated graphs with written explanations — demonstrating full progression from the opening timer-race puzzle to confident mathematical description of linear motion.
Supporting Phenomena	– Speed Camera Paradox (Road Safety): A video clip shows two vehicles on Thika Superhighway — one travelling at 80 km/h towards Nairobi and another at 80 km/h away from Nairobi. The speed camera records the same value for both, yet a head-on collision between them involves a closing speed of 160 km/h. Why does the direction matter so much? This supports the distinction between speed (scalar) and velocity (vector) and introduces relative speed. – SGR Train Motion Graph: Learners are shown a distance–time data table for a Madaraka Express journey from Nairobi to Mombasa, including station stops. Plotting this data produces a graph with flat sections (stationary at stations) and sloped sections (moving) — a puzzling staircase shape that motivates interpretation of displacement–time graphs. – Eliud Kipchoge's Sub-2-Hour Marathon: Learners examine split-time data from Kipchoge's INEOS 1:59 Challenge and

	calculate his average speed and velocity at different stages. The realisation that his velocity is not constant — he accelerates and decelerates — introduces the need for velocity–time graphs and the concept of acceleration. – Boda-Boda Braking Distance Illustration: A short illustrated scenario shows a boda-boda rider in Kisumu travelling at different speeds when a child runs onto the road. Learners compare braking distances at 30 km/h vs. 60 km/h and are shocked that doubling speed quadruples stopping distance — motivating the deep study of velocity–time graphs and the area-under-graph concept.

LESSON 1 (40 minutes): The Timer Race Puzzle: Launching Linear Motion	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor learners in a concrete, puzzling experience that reveals why direction matters in describing motion, and to launch the unit-long inquiry into linear motion.
Knowledge	Learners will begin to distinguish between distance travelled and final position, recognising that the same numerical distance can correspond to very different physical outcomes depending on path and direction.
Skills	Learners will practise observing, recording basic distance and time data, identifying anomalies in data, and formulating questions based on evidence.
Attitudes	Learners will develop curiosity and intellectual honesty by confronting a genuinely puzzling result and resisting premature closure — sitting comfortably with not-yet-answered questions.
Key Inquiry Question	How do distance, displacement, speed, and velocity differ from one another, and how are they calculated?
Purpose in Storyline	This is the opening lesson that launches the anchoring phenomenon. The Timer Race Puzzle creates the cognitive conflict that motivates every subsequent lesson. Learners will return to this experience repeatedly as they build richer mathematical tools to explain what they observed today.
Safety Notes	Conduct the outdoor activity in a clearly marked, obstacle-free section of the school compound. Brief learners to walk briskly — not run — to avoid falls. Ensure no learner has a physical condition preventing participation; assign an observer or recorder role for any such learner.

B. LESSON OVERVIEW

Learners go to the school compound and participate in a structured Timer Race in three groups. Group A walks a straight-line path from start to finish. Group B walks a winding zigzag path around cones or chalk markers and arrives at the finish in roughly the same time. Group C walks the zigzag path and then retraces steps back to a midpoint, also covering a similar total distance. Back in the classroom, learners compare their data and confront the puzzle: same distance, same time — yet the three groups end up in completely different locations. This launches the Driving Question Board and sets the agenda for the entire unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Introduction to harmonic motion] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Before going outside, learners sit in their groups and receive a hand-drawn map (a simple sketch on an A4 sheet) showing the school compound with a Start flag and a Finish flag. The map shows two routes: a straight arrow labelled Route A and a winding zigzag	Teacher distributes the map sketches and says: 'Before we go outside, I want you to think carefully about what you expect to happen. There are no wrong answers right now — just your best thinking.' Teacher writes the three prompt questions on the	Prediction elicitation using a visual map prompt activates prior intuitions about distance and position. Recording predictions publicly creates accountability and gives learners a concrete claim to test — a core sense-making move	Teacher scans written predictions in exercise books during circulation. Key diagnostic: do learners conflate distance with final position? Do any learners already distinguish path length from displacement? This informs the depth of questioning needed in

📖 **HTML:** [Absolute Value Distance; Graphs of Absolute Value Equations (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	arrow labelled Route B. Learners are told that two people travel both routes in the same time. In their exercise books learners respond individually to three prompt questions written on the chalkboard: (1) Which person do you think travels a greater distance? (2) Which person do you think is farther from the starting point when they finish? (3) If a third person walks Route B but turns back halfway, where do you predict they will end up? Learners share predictions with a partner, then two or three volunteers share with the class. Teacher records predicted answers in a two-column table on the board labelled OUR PREDICTION and WHAT WE FOUND OUT — the second column is left blank for now.	board slowly, reading each aloud. After individual thinking time (2 minutes of silence — teacher circulates without commenting), teacher says: 'Now turn to your neighbour. Share your predictions. Do you agree or disagree? Why?' WAIT TIME: Allow 90 seconds of pair talk. Teacher then cold-calls two or three pairs and records their predictions on the board without evaluation, saying only: 'Interesting — let us hold that idea and test it outside.'	that primes observation.	the Explain phase.
Observe Phase 📹 **VIDEO:** [Introduction to harmonic motion] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Absolute Value Distance; Graphs of Absolute Value Equations (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The class moves to the school compound. Teacher or a senior learner has already set up the course: a Start cone, a Finish cone approximately 20 metres away in a straight line, and four intermediate cones arranged in a zigzag pattern. Total zigzag path length is measured with a tape measure to be approximately 30 metres, and this figure is written on a card that learners can see. Group A (4 learners) walks the straight 20-metre line. Group B (4 learners) walks the 30-metre zigzag and is timed to arrive at Finish at roughly the same moment as Group A by starting slightly earlier or walking at a brisk pace (teacher adjusts timing through a countdown). Group C (2 learners) walks the zigzag but at the halfway cone turns around and walks back to a point clearly not at the Finish. The rest of the class observes and	Before the activity begins, teacher briefs: 'Your job is to be scientists. Watch carefully where each group ends up and record the data in your table. Do not just write what you expected — write what you actually see.' During the activity teacher narrates minimally, only directing attention: 'Watch where Group C stops. Note that down.' After all three groups have completed their routes, teacher gathers the class in a semicircle and poses: 'Look at where everyone is standing. What do you notice?' WAIT TIME: 30 seconds of silent looking before accepting any responses. Teacher then says: 'Turn and tell your partner two things you observed that surprised you.' WAIT TIME: 60 seconds. Teacher collects two or three observations aloud, scribing key phrases on a clipboard to bring back inside.	Firsthand observation in the school compound makes the phenomenon real and visceral. The freeze-in-place moment — where learners can literally see three people in three different locations — creates the visual anchor for the distance-versus-displacement distinction. Recording in a data table builds the habit of evidence-based reasoning.	Teacher checks that learners have completed their data tables with at least an approximate distance figure and a description of final position. The quality of their surprise comments reveals whether cognitive conflict has been genuinely triggered — a prerequisite for productive inquiry in subsequent lessons.

	records in a simple data table in their exercise books with columns: Group, Path Type, Approximate Distance Covered, Final Position (relative to Finish cone). After observing all three groups, learners note the striking result: Groups A and B are at the Finish, Group C is somewhere in the middle — yet Group C also covered significant distance. Teacher asks all groups to freeze at their final positions so the class can clearly see the spatial difference.			
Explain Phase 📹 **VIDEO:** Introduction to harmonic motion Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Absolute Value Distance; Graphs of Absolute Value Equations (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Back in the classroom, learners spend 8 minutes in groups of four attempting to explain the puzzle using only the vocabulary they currently possess. Each group is given a large sheet of paper (or uses the back of a manila card if paper is limited) and writes or draws their best explanation for: Why did Group C end up in a different place even though they covered a lot of distance? Groups may use diagrams, arrows, or words. After 5 minutes of group work, one representative from each group does a gallery share — standing beside their paper and giving a 30-second explanation. The class listens without interrupting. Teacher then poses three focusing questions to the whole class: (1) Is distance the same as where you end up? (2) What extra information would you need to fully describe where someone is after a journey? (3) Why might a matatu driver or a pilot need to know more than just distance? Learners respond briefly in their exercise books before a short whole-class discussion. Teacher introduces — without full	Teacher launches group work with: 'You have seen something puzzling. Your group has 5 minutes to write your best explanation on this paper. Use diagrams if they help. You do not need to be right — you need to think carefully.' During group work teacher circulates, listening for emerging ideas. Teacher avoids correcting misconceptions at this stage, instead asking probing questions: 'What if Group C kept walking the zigzag all the way to the finish — would they end up in the same place as Group A?' and 'What information is missing from the word distance?' After gallery share, teacher says: 'I am hearing some really important ideas about direction. There is a mathematical word for the shortest straight-line distance from start to finish that takes direction into account. We will build that idea fully in our next lesson. For now, write the word displacement in your book and put a question mark next to it.' WAIT TIME: After each focusing question, teacher waits a full 20 seconds before accepting responses, scanning the room to	Group explanation attempts — before formal instruction — surface existing mental models and force learners to articulate their thinking in their own language. The gallery share builds a shared classroom vocabulary that teacher can later refine. Introducing vocabulary labels without full definitions sustains productive tension and motivates Lesson 2.	Teacher listens during group work for the presence or absence of directional language. Can learners articulate that Group C moved in two directions? Are they already using words like back, return, or opposite? This formative data shapes the depth of vocabulary instruction in Lesson 2. Teacher notes two or three groups to cold-call in the next lesson.

	definition yet — the words displacement and direction as words the class will investigate in Lesson 2, writing them on the DQB wall space.	ensure most learners have had thinking time.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Introduction to harmonic motion] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Absolute Value Distance; Graphs of Absolute Value Equations (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or two small squares of paper with tape if sticky notes are unavailable). On the first note, learners write one thing they are now WONDERING about motion — a genuine question prompted by the race activity. On the second note, learners write one thing they think they already KNOW about motion. Teacher collects the know notes in a basket for later analysis and invites learners one by one (or in a quick pass-around) to place their wonder notes on a dedicated section of the classroom noticeboard or wall labelled DRIVING QUESTION BOARD — Our Questions About Motion. Teacher reads five or six of the questions aloud as they are posted, grouping similar questions together with a quick gesture. Example questions that typically emerge: Why does direction change where you end up? What is the difference between speed and velocity? How do you calculate how fast something is moving? Why does a matatu speedometer not show direction? Could we make a graph of the race? Teacher highlights the unit Driving Question written on a banner above the board: How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter? Teacher says: 'Every question on this board is a real mathematical question. By	Teacher frames the DQB activity with: 'Scientists and mathematicians keep a record of what puzzles them. Your wonder questions are not signs that you do not know enough — they are the engine of this unit. Write your most honest, most curious question.' Teacher circulates during writing (90 seconds of silent individual writing). As learners post their notes, teacher narrates lightly: 'I can see questions about graphs, about speed, about direction — these are exactly the ideas mathematicians who study motion have grappled with.' Teacher deliberately does NOT answer any question at this stage. If a learner asks for an answer, teacher responds: 'That is a perfect question for our board. Let us hold it there and find the mathematics to answer it.' Teacher points to two or three questions and says: 'We will come back to read these again at the start of Lesson 2.'	The DQB externalises learner curiosity and creates a living record of the inquiry. Grouping similar questions makes the intellectual terrain of the unit visible. Referencing the DQB in subsequent lessons creates continuity and helps learners experience themselves as investigators whose questions drive the curriculum.	Teacher reads the know notes after class to diagnose baseline understanding. Key diagnostics: Do learners confuse distance and displacement? Do any already know the term velocity? Are there misconceptions about speed being always positive or always in one direction? This informs pacing decisions for Lessons 2 and 3.

	Lesson 10 we will have answered all of them using mathematics.'			
Model Building Phase 📹 **VIDEO:** [Introduction to harmonic motion] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Absolute Value Distance; Graphs of Absolute Value Equations (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, learners individually draw a simple initial model in their exercise books — a sketch of the school compound showing the Start and Finish points and the three paths taken by the three groups. They annotate their sketch with labels and short sentences answering: What information does the number 30 metres give us? What information does it NOT give us? This is an initial model — intentionally incomplete and personal. Teacher tells learners: 'This sketch is your starting point. We will add to it, revise it, and make it more mathematically precise in every lesson of this unit. By Lesson 10 your model will include graphs, equations, and a full mathematical language for motion.' Learners write the date and label the sketch INITIAL MODEL — Lesson 1 so they can revisit it.	Teacher explains the model task with: 'A model is not just a picture — it is a thinking tool. Your sketch should show what happened and begin to show what questions remain. Use arrows to show direction of travel for each group.' Teacher circulates and prompts with: 'Can you add an arrow showing which direction Group C was facing at the end?' and 'Can you write one question your model cannot yet answer?' In the final minute teacher closes with: 'Next lesson we will watch two short video clips that will give us the first mathematical tools to start answering the questions on our board. Come ready to think about the words distance, displacement, speed, and velocity.' Teacher points to the DQB: 'These questions will start getting answers very soon.'	The initial model establishes the practice of model building that runs through all ten lessons. Labelling it INITIAL MODEL signals to learners that revision is expected and desirable — a key epistemic practice of science and mathematics. Connecting forward to Lesson 2 maintains narrative momentum and links the phenomenon to the unit storyline.	Teacher scans models during circulation. Key checks: Has the learner shown three distinct paths? Has the learner used directional arrows? Has the learner identified at least one piece of missing information? These models can be photographed or reviewed after class to plan targeted support in Lesson 2.

D. TEACHER REFLECTION

After the lesson, consider: Did genuine cognitive conflict occur — did learners seem genuinely puzzled rather than indifferent? Were the three groups spatially distinct enough for the contrast to be obvious? Did the DQB questions reveal useful baseline knowledge and misconceptions? Which groups articulated directional thinking unprompted, and which still conflated distance with position? How will you sequence the Lesson 2 video clips to best address the most common misconception observed today? Note two or three specific learner questions from the DQB to reference explicitly in Lesson 2 to reinforce continuity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two groups covering different path lengths arrived at the same finish point at the same time, while a third group that retraced part of the zigzag path ended up at a completely different location despite also covering significant distance.
What did I learn?	The total length of a path and the final position of a traveller are not the same thing. The number representing distance alone does not tell us where someone ends up — direction of travel matters.

How does this explain the phenomenon?	Learners attempted to explain the puzzle using their own language, identifying that moving in two opposite directions can cancel out progress toward the finish. The word displacement was introduced as the mathematical term the class will define precisely in Lesson 2.

LESSON 2 (40 minutes): Distance, Displacement, Speed and Velocity: Giving Direction to Numbers	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will reconcile their timer-race observations with video evidence to formally define and distinguish distance from displacement and speed from velocity, and will apply these definitions to solve introductory calculations using Kenyan contexts.
Knowledge	Learners will define distance, displacement, speed, and velocity; distinguish scalar quantities from vector quantities; and state the formulae for average speed and average velocity.
Skills	Learners will calculate average speed and average velocity from given data, represent displacement as a directed quantity on a number line, and interpret short video clips as evidence for scientific definitions.
Attitudes	Learners will appreciate that precision in mathematical language reflects real-world accuracy, showing curiosity when comparing their own timer-race experience with the formal definitions.
Key Inquiry Question	How do distance, displacement, speed, and velocity differ from one another, and how are they calculated?
Purpose in Storyline	Lesson 1 created cognitive conflict: same distance, different positions. Lesson 2 resolves part of that conflict by giving learners the formal vocabulary and formulae for distance, displacement, speed, and velocity, building the mathematical language needed to re-examine the timer-race puzzle and to tackle acceleration, graphs, and relative speed in later lessons.
Safety Notes	No physical activity in this lesson. If video is projected, ensure the room lighting is adequate and screen glare is minimised. Learners with visual impairments should be seated closest to the screen.

B. LESSON OVERVIEW
The lesson opens by revisiting the timer-race puzzle through a brief prediction task that surfaces prior thinking. Learners then watch two short video clips — the first on distance versus displacement, the second on speed versus velocity — pausing after each to discuss and reconcile the video evidence with their personal timer-race experience. In the Explain phase, the class co-constructs formal definitions and formulae, then solves two worked examples set in Kenyan contexts (a cyclist travelling from Kericho to Nakuru; a matatu making a return trip along Thika Road). The lesson closes with learners updating the Driving Question Board and sketching a simple arrow model on a number line to represent displacement.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Finding Total Distance Using Ratios Source: CK-12 🔍 Search ARES	Learners open their exercise books and respond individually to the prompt displayed on the board: 'In the timer race yesterday, Group A walked 80 m in a straight line and Group B walked 80 m along a zigzag path. Both took 40	Display the prompt and say: 'Do not worry if you have never heard the word velocity before. Use the letters in the word, what you remember from yesterday, or any science you already know. I want your honest prediction, not the	Elicitation of prior knowledge and prediction journaling. Learners activate memory of the timer-race phenomenon and make a falsifiable prediction that the video evidence will either confirm or challenge, creating personal	Circulate during the 90-second writing and note whether learners conflate speed with velocity or show any awareness of direction. Use these observations to target cold-call questions. Listen for the word direction — if no learner uses

for similar videos 📖 **HTML:** Absolute Value Distance; Graphs of Absolute Value Equations (PLIX) Source: CK-12 🔍 Search ARES for similar readings	seconds. Write one sentence predicting: do the two groups have the same speed? Do they have the same velocity? You may not yet know the word velocity — write your best guess about what the word might mean, based on the root of the word.' Learners write for 90 seconds, then share with a shoulder partner for 60 seconds.	right answer.' Start a 90-second silent writing timer. After writing, say: 'Now share your sentence with the person next to you. You have 60 seconds. Listen carefully — you will need to tell the class what your partner said.' After paired sharing, cold-call three pairs and ask: 'What did your partner predict, and do you agree?' WAIT TIME: After each pair responds, pause for 5 seconds before commenting, allowing other learners to react first. Avoid confirming or correcting — say: 'Interesting — let us hold that idea and see what the video shows us.'	investment in the observation phase.	it, note this to highlight after the first video clip.
Observe Phase 📹 **VIDEO:** Finding Total Distance Using Ratios Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Absolute Value Distance; Graphs of Absolute Value Equations (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners watch Video Clip 1 (approximately 3 minutes) on distance versus displacement, showing a figure moving from point A to point B along two different paths. After the clip pauses, learners respond in their books: 'What was the key difference the video showed between distance and displacement?' They then watch Video Clip 2 (approximately 3 minutes) on speed versus velocity, which uses a matatu scenario. After the second clip, learners answer: 'According to the video, what one extra piece of information turns a speed into a velocity?' Learners then compare their written responses with the prediction they wrote in the Predict phase.	Before playing Clip 1, say: 'As you watch, keep your timer-race experience in mind. At what moment does the video remind you of something you noticed yesterday? I will pause after each clip and ask you to write before we discuss.' Play Clip 1. PAUSE. Say: 'Write your answer now — one sentence. You have 45 seconds. No talking yet.' After writing, ask: 'Who noticed something in the video that matched what you experienced in the timer race?' WAIT TIME: 5 seconds after the question before taking responses. Take two or three responses, then play Clip 2. PAUSE again. Say: 'Write your answer about speed and velocity now. 45 seconds.' After writing, ask: 'How does the matatu in the video connect to the Nairobi matatus we know? Could a matatu travelling at 80 km/h tell you which direction Nairobi is?' WAIT TIME: 5 seconds. Accept two responses, then say: 'Let us now use what we have seen to build proper definitions together.'	Video-based evidence gathering with guided written note-taking. By pausing and writing before discussing, learners process video content individually first, preventing groupthink and giving the teacher formative information about comprehension before whole-class discussion begins.	Collect written responses by asking three learners to read aloud. Listen for: (1) use of the word direction when defining displacement; (2) understanding that velocity requires a stated direction. If learners describe speed and velocity as identical, note this as a target for the Explain phase worked example.

Explain Phase 📹 **VIDEO:** Finding Total Distance Using Ratios Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Absolute Value Distance; Graphs of Absolute Value Equations (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The class co-constructs a definitions table in their books. The teacher guides them to complete four rows: Distance (total path length, scalar, unit: metres), Displacement (straight-line distance from start to finish with direction stated, vector, unit: metres with a direction), Speed (distance divided by time, scalar, unit: m/s or km/h), Velocity (displacement divided by time with direction, vector, unit: m/s or km/h with direction). Learners then solve two worked examples. Example 1: A cyclist leaves Kericho and rides 120 km north-east to Nakuru in 3 hours. Calculate the average speed and average velocity. Example 2: A matatu travels 30 km from Thika to Nairobi, then 30 km back to Thika in a total of 2 hours. Calculate the average speed and average displacement for the whole journey. Learners attempt each example individually for 2 minutes before the teacher works through it on the board with class input.	Draw a two-column table on the board labelled Scalar and Vector. Say: 'Based on the videos and your timer-race experience, help me fill this in. Which quantities need a direction to be fully described, and which do not?' WAIT TIME: 5 seconds after asking. Take learner suggestions. Then say: 'Let us write the formal definition together.' Dictate the four-row table slowly, asking learners to copy. For Example 1, say: 'Before I write anything on the board, attempt this yourself for 2 minutes. Use the formula: speed = distance divided by time.' After 2 minutes, say: 'Who has an answer for average speed? Who has an answer for average velocity? Are they the same number? Can they be different numbers for the same journey?' WAIT TIME: 5 seconds after each question. For Example 2, prompt: 'What is the total distance? What is the displacement? Discuss with your partner for 30 seconds before you answer.' After Example 2, ask: 'Why is the average velocity zero even though the matatu was moving the whole time? Connect your answer back to the timer-race puzzle.' WAIT TIME: 7 seconds — this is a high-order question. If no response, say: 'Think about Group C from yesterday who retraced their steps. What was their final position?'	Collaborative definition-building followed by worked examples with deliberate connection back to the anchoring phenomenon. The matatu return-trip problem mirrors the timer-race puzzle exactly, helping learners see that the cognitive conflict from Lesson 1 is resolved by the concept of displacement and velocity rather than distance and speed.	As learners attempt examples independently, circulate and check: (1) whether learners correctly identify displacement as zero for the return trip; (2) whether velocity is given with a direction for Example 1. Use targeted questioning for learners who write identical values for speed and velocity in Example 2. Ask them: 'What is the starting point? What is the ending point? How far apart are they?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** DNA as the "transforming principle"	Learners return to the Driving Question Board created in Lesson 1. Working in pairs, they choose one sticky note from the DQB that their new learning today can partially or fully answer. They write a revised answer or a new label on a different-coloured sticky note	Say: 'Yesterday you posted questions that puzzled you. Today we have new evidence from the videos and calculations. Walk to the DQB and find a question your group can now partially answer with what you learned today.' Give pairs 2 minutes to browse, then 2	DQB revision as evidence tracking. Learners physically move between answered and unanswered question zones, making the storyline progression visible and reinforcing the idea that science and mathematics build incrementally. New questions	The revised sticky notes serve as a formative evidence artefact. Collect photographs of the DQB at the end of the lesson. Review whether learners correctly apply displacement or velocity language in their revised answers. Note any persistent misconceptions (for

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Absolute Value Distance; Graphs of Absolute Value Equations (PLIX) Source: CK-12 🔍 Search ARES for similar readings	and attach it beside the original question. They also add at least one new question prompted by today's lesson — a question about what they still do not understand or what they are now curious about (for example, questions about acceleration or about how to show velocity on a graph).	minutes to write and post their revised answer and new question. Circulate and read the new sticky notes aloud to the class, grouping emerging questions about graphs and acceleration on one side of the board. Say: 'Notice how answering one question opens up new questions. The new cluster forming on the right side of the board — about how we draw velocity or how velocity changes — is exactly where we are heading in Lessons 3 and beyond.' WAIT TIME: After reading each new question aloud, pause 4 seconds and ask: 'Does anyone else share this question? Raise your hand.' This validates learners who had the same unresolved confusion.	about acceleration and graphs set up the need for Lessons 3 through 8.	example, learners who still equate distance with displacement in their revised notes) for targeted follow-up at the start of Lesson 3.
Model Building Phase 📹 **VIDEO:** Finding Total Distance Using Ratios Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Absolute Value Distance; Graphs of Absolute Value Equations (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners draw a simple number-line model in their exercise books. They mark a starting point (0 m), an intermediate point (representing the zigzag midpoint from the timer race), and an ending point. They draw one arrow from 0 to the end showing displacement of the straight-line walker, and a second arrow showing the net displacement of the retracing walker (who ends up back at the midpoint). They label each arrow with a direction word (north, east, or left/right) and a magnitude, then write one sentence below the diagram: 'The arrows in my model show that displacement is different from distance because ...' Learners share their sentence with their partner before the lesson ends.	Say: 'We are going to make a simple arrow diagram — a model — that captures everything we learned today in one picture. Models in mathematics and science help us see patterns quickly. Draw a horizontal number line. Mark your start, your midpoint, and your end.' Demonstrate on the board as learners draw. Say: 'Now draw one arrow for the straight-line walker and one for the retracing walker. Use direction words you know — left, right, north, south, east, west — your choice.' After 3 minutes of individual drawing, say: 'Finish your sentence: The arrows in my model show that displacement is different from distance because ...' WAIT TIME: After learners write, ask two volunteers to read their sentences aloud. Pause 5 seconds after each. Then say: 'Your arrow models are early versions of something much more powerful — displacement-time graphs —	Arrow-diagram model building as a bridge between concrete experience and abstract graphical representation. The number-line model is deliberately simple so all learners can succeed, yet it encodes the key conceptual distinction between scalar distance and vector displacement. Naming it a model and linking it explicitly to future graph lessons advances the storyline and builds anticipation.	Examine the arrow diagrams and sentences as learners work. Key success indicators: arrows have both a length (magnitude) and a direction label; the retracing walker arrow ends at a different position from the straight-line walker arrow; the sentence uses the word direction or position. Learners who draw arrows of equal length for both walkers need a prompt: 'Where does each person end up standing? Mark that position, then draw the arrow to there.'

		which we will build in Lesson 5. Keep this page; you will need it again.'		

D. TEACHER REFLECTION
After the lesson, consider: Did the video clips generate genuine discussion or did learners passively watch? Did the Kericho-to-Nakuru and matatu return-trip examples feel real to learners or abstract? Were learners able to connect the zero-displacement result of the return trip back to the timer-race retracing walker without being told? Which questions on the DQB suggest readiness for acceleration in Lesson 3, and which suggest unresolved confusion about the speed-velocity distinction that needs brief revisiting at the start of Lesson 3? Note the names of any learners whose arrow models showed persistent conflation of distance and displacement for targeted support.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We watched two video clips showing that a person can travel a long path but end up close to where they started, and that the speedometer of a matatu does not tell you which direction the matatu is heading.
What did I learn?	Distance is the total length of the path travelled (scalar). Displacement is the straight-line distance from start to finish with direction stated (vector). Speed equals distance divided by time. Velocity equals displacement divided by time and must include direction. A return journey to the starting point has zero displacement and therefore zero average velocity, even though average speed is not zero.
How does this explain the phenomenon?	The timer-race puzzle is explained by the difference between displacement and distance. Group C had the same distance as the straight-line walker but a smaller displacement because they retraced steps, changing their direction and therefore their final position. The number on its own — whether distance or speed — does not tell us where someone ends up or which way they are going. Direction always matters.

LESSON 3 (80 minutes): Velocity, Speed & Acceleration: Illustrations + Calculations	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive the formula for acceleration, distinguish between speed and velocity, and calculate acceleration and deceleration in real-world Kenyan motion contexts.
Knowledge	Learners will know that: (1) velocity is the rate of change of displacement with time and is a vector quantity; (2) acceleration is the rate of change of velocity with time, expressed as $a = (v - u) / t$; (3) positive acceleration means speeding up, negative acceleration (deceleration) means slowing down, and zero acceleration means constant velocity.
Skills	Learners will be able to: (1) calculate velocity using v = displacement / time; (2) derive and apply the formula $a = (v - u) / t$ to solve multi-step problems; (3) distinguish between scalar speed and vector velocity using direction; (4) identify and interpret positive, negative, and zero acceleration from given data.
Attitudes	Learners will appreciate the importance of mathematical precision in describing real-world motion, and develop confidence in multi-step problem solving using familiar Kenyan transport contexts.
Key Inquiry Question	How do distance, displacement, speed, velocity, and acceleration differ from one another, and how are they calculated?
Purpose in Storyline	Building on Lessons 1 and 2, where learners established the difference between distance and displacement and between speed and velocity, this lesson deepens that understanding by introducing acceleration — the rate at which velocity changes. Using illustrated scenarios of a matatu accelerating from a Nairobi bus stop and a motorcycle braking along Thika Road, learners discover why knowing speed alone is still not enough: we also need to know how quickly velocity is changing and in which direction. This advances the driving question by equipping learners with the full trio of motion quantities — velocity, acceleration, and direction — needed to describe and predict straight-line motion.
Safety Notes	No physical activity outdoors in this lesson. All scenarios are illustration-based. Ensure learners do not run or move furniture during the role-play segment. If a calculator is used, remind learners to check units before substituting values.

B. LESSON OVERVIEW
Learners open by revisiting the Timer Race Puzzle phenomenon and the Driving Question Board from Lessons 1–2. They then examine two illustrated scenarios — a matatu accelerating away from a Nairobi CBD bus stop and a boda-boda motorcycle braking before a speed bump on Ngong Road — to gather new evidence. Through guided sensemaking, they derive the acceleration formula $a = (v - u) / t$, distinguish positive from negative acceleration, and complete a structured set of multi-step calculations in pairs. The lesson closes with DQB revision and a model-building sketch that adds acceleration arrows to their existing displacement-velocity diagrams.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [AP Physics 1]()	Learners are shown two illustrated panels side by side on the board (or printed handout): Panel A	Display the two illustrated panels and read the scenario aloud. Say: 'Before we do any mathematics, I	Think-Pair-Share with Prediction Column: Learners externalise prior reasoning before instruction,	Observable indicator: All learners have written at least one sentence per prompt before pair talk begins.

review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	shows a matatu at a Nairobi CBD bus stop with passengers boarding, its speedometer reading 0 km/h; five seconds later it reads 60 km/h. Panel B shows a boda-boda motorcycle at 80 km/h approaching a speed bump on Ngong Road; three seconds later it reads 20 km/h. Without any formula, learners individually write predictions in their exercise books answering three prompts: (1) 'Which vehicle's velocity changed more in total?' (2) 'Which vehicle's velocity changed faster?' (3) 'Is the motorcycle's change in velocity the same kind of change as the matatu's? Explain.' Learners then share predictions with a partner (Think-Pair) before the teacher cold-calls 4 pairs.	want you to think like a mathematician. Look at these two vehicles and write your three predictions — you have 3 minutes, working alone.' Start a visible 3-minute timer. After individual writing, say: 'Now turn to your partner and compare what you wrote. Do you agree? Where do you disagree?' Allow 2 minutes of pair talk. After pair talk, cold-call exactly 4 pairs using a random name stick: 'Tell me your answer to prompt 2 — which vehicle's velocity changed faster, and how did you decide?' Apply 10–12 seconds of WAIT TIME after each cold-call before probing. Record key prediction words on the board in a 'Prediction Column' (e.g., 'bigger change', 'faster change', 'slowing down is different'). Do NOT confirm or deny any prediction yet. Close with: 'Hold these predictions — our evidence activity will help us test them.'	activating intuitive ideas about rates of change. The Prediction Column on the board creates a public record that will be revisited and revised in Phase 3, making thinking visible and motivating evidence-gathering.	Listen during cold-calls for whether learners spontaneously use directional language ('slowing down', 'opposite direction') — this reveals carry-over from Lessons 1–2. Flag any learner who conflates speed with velocity for targeted follow-up in Phase 3.
Observe Phase 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a structured illustrated worksheet titled 'Two Journeys on Nairobi Roads'. The worksheet contains: (A) a detailed annotated diagram of the matatu scenario with a data table (initial velocity u = 0 m/s, final velocity v = 16.7 m/s, time t = 5 s); (B) a detailed annotated diagram of the boda-boda scenario with a data table (u = 22.2 m/s, v = 5.6 m/s, t = 3 s); (C) a blank 'What I Notice' column alongside each diagram. Learners study both diagrams silently for 4 minutes, completing the 'What I Notice' column. They then complete a guided data-extraction table: recording the change in velocity (v − u) for each vehicle, computing the ratio (v − u) / t for each, and recording whether	Distribute the worksheet and say: 'You have 4 minutes to study both diagrams silently and fill in your 'What I Notice' column. I am not answering content questions yet — just observe.' Circulate and look over shoulders; do not correct, only prompt observation with: 'What do the numbers in the data table tell you?' After 4 minutes, say: 'Now move to the data-extraction table. Work through it step by step. You have 6 minutes.' Circulate and pause at tables where learners are subtracting in the wrong order (v − u vs. u − v). Ask: 'Which velocity came first in time? So which do we subtract from which?' Apply 12 seconds WAIT TIME. After 6 minutes, cold-call 3 learners to read out their (v −	Structured Data Extraction with Annotated Diagrams: Rather than presenting the formula directly, learners extract the mathematical structure (v − u) / t from real data, allowing the formula to emerge from evidence. The 'What I Notice' column builds observational discipline (NGSS Practice 3: Planning and Carrying Out Investigations; Practice 4: Analysing and Interpreting Data).	Observable indicator: Learners correctly identify v − u as negative for the boda-boda and positive for the matatu. Check that learners write units (m/s²) next to computed ratios. A correct bridging answer reads something like: 'A negative value means the velocity is decreasing — the boda-boda is slowing down.' Collect 5 random worksheets at the end of this phase to check subtraction order and unit use before Phase 3.

	the change is positive or negative. Finally, learners answer a bridging question: 'What does a negative value of (v − u) / t tell you about the motion of the boda-boda?'	u) / t values and whether positive or negative. Write both values on the board without labelling them as 'acceleration' yet. Ask the class: 'What single word might describe what this ratio is measuring?' Wait 10 seconds. Accept all vocabulary suggestions and write them beside the values.		
Explain Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher formalises the formula a = (v − u) / t, connecting it directly to learners' computed ratios. Learners copy the definition, formula, and three cases (positive, negative, zero acceleration) into their notes using a provided graphic organiser. They then work in pairs through four graduated calculation problems, all set in Kenyan contexts: (1) A Kisumu ferry accelerates from rest to 8 m/s in 10 s — find acceleration. (2) A Nairobi Express bus slows from 25 m/s to 10 m/s in 5 s — find acceleration and state whether it is deceleration. (3) A Kipchoge-inspired marathon runner maintains exactly 5.83 m/s for 2 km — state the acceleration. (4) A challenge problem: a matatu travelling at 72 km/h brakes and stops in 8 s — convert units, then find the deceleration. After pair work, learners return to the Prediction Column on the board and vote (thumbs) on which predictions were confirmed, partly confirmed, or overturned.	Begin by pointing to the two (v − u) / t values on the board from Phase 2 and say: 'You have already calculated acceleration — you just didn't have the name for it yet. The rate of change of velocity with time is called acceleration.' Write formally: a = (v − u) / t. Explain each variable with 15 seconds per variable, asking: 'What does u stand for? What does v stand for?' Cold-call 2 learners each time with 10 seconds WAIT TIME. Distribute the graphic organiser and give learners 3 minutes to complete it independently. Then say: 'Work in pairs on the four problems. For each one, write: what is known, what is unknown, the formula, substitution, and the answer with units. You have 15 minutes.' Circulate. At problem 4, prompt unit conversion by asking: 'Your velocity is in km/h — can you substitute that directly into a formula that gives m/s²? What do you need to do first?' Wait 12 seconds before guiding. After pair work, facilitate a 5-minute whole-class Prediction Column revisit: point to each original prediction and ask the class to vote. Say: 'Your prediction said [quote prediction]. Now that you have evidence and the formula, was this confirmed?' Cold-call 3 learners to justify the vote with mathematics.	Formula Emergence + Graduated Problem Scaffolding: By naming the formula only after learners have already computed it from data, the formula becomes a label for a structure they already understand rather than an abstract rule. Graduated problems (rest-to-motion, motion-to-rest, constant velocity, unit conversion) build procedural fluency alongside conceptual understanding (NGSS Practice 5: Using Mathematics and Computational Thinking). The Prediction Column revisit closes the epistemic loop between prediction and evidence.	Observable indicator: In problem 2, learners must write a negative value (a = −3 m/s²) AND label it as deceleration. In problem 3, learners must write a = 0 m/s² and explain why. In problem 4, learners must show a unit-conversion step (72 km/h ÷ 3.6 = 20 m/s) before substituting. Collect pair-work sheets. Any learner who writes a positive value for problem 2, skips units, or substitutes 72 directly for v in problem 4 requires immediate follow-up via a targeted exit prompt.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class DQB (a large chart or pinboard at the front). They individually write on sticky notes: (1) one thing they can now answer from the driving question that they could not answer before this lesson (GREEN note); (2) one new question this lesson has raised for them (YELLOW note). Learners place their notes on the DQB and the teacher reads out a selection aloud, clustering related questions together. The class identifies which DQB questions from Lessons 1 and 2 can now be answered using today's knowledge of acceleration, and they mark those with a tick.	Direct learners to the DQB and say: 'Take two sticky notes. On the green one, write one specific thing you can now answer about our driving question — be precise, use the word velocity or acceleration. On the yellow one, write a new question today's lesson has created for you. You have 3 minutes.' Circulate and prompt vague green notes: 'Can you write a formula or a number to make that more specific?' After 3 minutes, read out 6–8 notes aloud (mix of green and yellow). Ask the class: 'Does anyone have a green note that answers one of the yellow notes from a previous lesson?' Cold-call 2–3 learners. Cluster yellow notes that ask about graphs of motion: 'Several of you are asking how we can see acceleration on a graph — that is exactly what Lesson 4 will address. Keep that question alive.' Leave the DQB updated and visible.	Driving Question Board Revision: The DQB makes the progression of the class's collective understanding visible over time. Colour-coding (green = answer, yellow = new question) distinguishes between consolidation and productive confusion, validating both as part of scientific inquiry (NGSS Practice 1: Asking Questions). Clustering emerging graph-related questions creates a conceptual bridge to Lesson 4.	Observable indicator: Green notes should reference acceleration, velocity, deceleration, or a formula — not vague statements like 'I learned about speed.' Yellow notes should be genuine questions, not restatements of content. The teacher reads all notes during collection and flags any learner whose green note shows a misconception (e.g., 'acceleration always means going faster') for a brief one-to-one check before the next phase.
Model Building Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative motion model diagram started in Lesson 1 (a sketch of the Timer Race Puzzle scenario with displacement and velocity arrows added in Lesson 2). Today, they add a third layer: acceleration arrows. Using the matatu scenario, they draw: (a) a rightward displacement arrow from the bus stop; (b) a rightward velocity arrow growing larger over 5 seconds; (c) a rightward acceleration arrow labelled $a = +3.34\ m/s^2$ to show speeding up. They then add the boda-boda scenario below: (a) rightward displacement arrow; (b) rightward velocity arrow shrinking over 3 seconds; (c) a leftward (or negatively labelled) acceleration	Say: 'Take out your cumulative model from Lesson 1. Today we are adding the third layer — acceleration arrows. Draw the matatu scenario first. Your acceleration arrow must have a direction AND a numerical label. You have 5 minutes.' Circulate and check that arrow directions are consistent with the sign of acceleration (a common error is drawing a leftward arrow for positive acceleration). Prompt: 'Your matatu is speeding up to the right — which way should the acceleration arrow point?' Wait 10 seconds. After 5 minutes, say: 'Now add the boda-boda. Remember — negative acceleration does NOT mean the	Cumulative Model Revision (Layered Diagram): Returning to and building on the same physical diagram across lessons makes conceptual progression tangible. Adding acceleration as a third visual layer — on top of displacement and velocity — reinforces that these are distinct but related quantities, all needed to fully answer the driving question (NGSS Practice 2: Developing and Using Models). The phenomenon annotation keeps the real-world puzzle at the centre of all model building.	Observable indicator: Learners correctly draw a rightward acceleration arrow for the matatu and an arrow opposing motion (or labelled negative) for the boda-boda. The phenomenon annotation must mention direction change or sign change in velocity — not just 'the person moved around.' Collect 6 models at random to check layering completeness before Lesson 4.

	arrow labelled a = −5.53 m/s² to show slowing down. Finally, learners annotate: 'In the Timer Race Puzzle, the zigzag walker who retraced steps had a changing direction — meaning their velocity changed sign — which means they experienced acceleration even without speeding up.' Learners share models with a neighbour for a 2-minute peer-review using the prompt: 'Is every arrow labelled with a value and a direction?'	arrow points to the left automatically; it means the acceleration opposes the direction of motion. Think carefully.' Allow 4 minutes. Then say: 'Write the annotation connecting your model back to the Timer Race Puzzle. One sentence is enough.' After annotation, instruct peer review: 'Show your neighbour your model. They must check: Does every arrow have a value? Does every arrow have a direction? You have 2 minutes.' Cold-call 2 pairs to share one thing their neighbour corrected.		

D. TEACHER REFLECTION
After this lesson, reflect on the following: (1) Did learners successfully derive the acceleration formula from the data before you named it, or did most pairs wait passively for you to provide it? If the latter, consider adding a more explicit 'notice the pattern' prompt to the worksheet in Phase 2. (2) In problem 4 (unit conversion), how many pairs converted correctly without prompting? If fewer than half did, plan a 10-minute unit-conversion warm-up at the start of Lesson 4. (3) Were learners' DQB green notes specific and formula-based, or vague and descriptive? Vague notes signal that the Explain phase did not produce enough consolidation — consider adding one more worked example in Lesson 4's opening. (4) Did any learner articulate in their model annotation that the zigzag walker who retraced steps 'accelerated' by changing direction? This is a sophisticated insight that bridges to the vector nature of velocity and should be celebrated publicly in Lesson 4's opening. (5) Check whether the boda-boda deceleration scenario raised questions about negative velocity (can something have a negative speed?) — if so, note this for the displacement-time graph work in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Two illustrated scenarios: a matatu accelerating from 0 to 60 km/h at a Nairobi CBD bus stop (Panel A) and a boda-boda braking from 80 km/h to 20 km/h before a speed bump on Ngong Road (Panel B). Data tables showing initial velocity (u), final velocity (v), and time (t) for each vehicle. Computed ratios (v − u) / t — one positive, one negative — extracted from real data before the formula was named.
What did I learn?	Velocity is the rate of change of displacement with time and is a vector quantity (has both magnitude and direction). Acceleration is the rate of change of velocity with time: a = (v − u) / t. Positive acceleration means the object is speeding up; negative acceleration (deceleration) means the object is slowing down; zero acceleration means constant velocity. Direction always matters — a boda-boda slowing down has a negative acceleration even though it is still moving forward. In the Timer Race Puzzle, the zigzag walker who retraced steps changed the direction of their velocity, which means they experienced acceleration — not just a change in position.
How does this explain the phenomenon?	Using the matatu and boda-boda illustrated scenarios, learners computed (v − u) / t from data before the teacher named it 'acceleration', allowing the formula to emerge from evidence rather than be imposed. Four graduated calculation problems (Kisumu ferry, Nairobi Express bus, marathon runner, unit-conversion matatu challenge) built procedural fluency and revealed that zero

	acceleration means constant velocity. The cumulative model diagram was updated with acceleration arrows, making the relationship between displacement, velocity, and acceleration visually explicit and anchored to the Timer Race Puzzle phenomenon.

LESSON 4 (80 minutes): Velocity and Acceleration Problems — Practice & Sensemaking	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To deepen learners' understanding of velocity and acceleration as vector quantities by solving structured problems in Kenyan contexts and justifying reasoning through group presentation.
Knowledge	Velocity is the rate of change of displacement with time (v = displacement ÷ time) and is a vector quantity. Acceleration is the rate of change of velocity with time (a = (v − u) ÷ t) and can be positive, negative (deceleration), or zero. Uniform acceleration means constant rate of change of velocity; non-uniform acceleration means a changing rate.
Skills	Calculate velocity and acceleration using appropriate formulae; distinguish between scalar and vector quantities in motion contexts; interpret the sign (positive/negative) of velocity and acceleration; present and justify mathematical solutions to peers.
Attitudes	Appreciate the precision that vector quantities bring to describing real-world motion; develop perseverance in multi-step problem solving; value collaborative reasoning and respectful critique of peers' solutions.
Key Inquiry Question	How do distance, displacement, speed, velocity, and acceleration differ from one another, and how are they calculated?
Purpose in Storyline	In Lessons 1–3 learners established what distance, displacement, speed, and velocity mean and why direction matters. Lesson 4 is the structured practice pivot: learners now apply velocity and acceleration formulae to rich Kenyan contexts — Eliud Kipchoge's marathon pace, a matatu braking on Thika Road, a fruit falling from a mango tree in Kisumu — and present their reasoning. This consolidates the mathematical language needed before graphical representation (Lessons 5–6) and builds the argument that direction is mathematically encoded in the sign of every vector answer.
Safety Notes	No physical movement or equipment required. Ensure group discussions remain respectful. If learners reference the Timer Race, remind them not to re-enact running in the classroom.

B. LESSON OVERVIEW

Learners open by revisiting the Driving Question Board and identifying the evidence gaps around velocity and acceleration. Working in structured groups of 4–5, they solve a tiered problem set drawn from Kenyan contexts (Eliud Kipchoge's marathon, a Nairobi matatu braking, a mango falling near Lake Victoria, and a boda-boda overtaking on Nakuru highway). Each group is assigned a different problem, solves it collaboratively using the formulae v = s/t and a = (v − u)/t, then presents their solution and justification to the class. Whole-class sensemaking follows, connecting every answer back to the anchoring phenomenon: the sign of the velocity or acceleration answer encodes direction — exactly the missing piece the Timer Race puzzle revealed. The lesson closes with a Driving Question Board update and an exit ticket.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [AP Physics 1]	Learners individually read two short scenarios — (A) Eliud Kipchoge runs 500 m east in 90 s,	Display both scenarios on the board or printed cards. Say: 'Before we do any mathematics, I	Prediction Anchoring — by committing to a prediction before instruction, learners activate prior	Observable indicator: Learner written predictions reference direction or sign in their reasoning

review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	then 200 m west in 40 s; (B) a matatu on Thika Road accelerates from 20 m/s to 35 m/s in 10 s — and write predictions: What do you think his average velocity is for the whole journey? What do you think the matatu's acceleration is? Will the sign (+ or −) of either answer matter? Learners record predictions in their exercise books BEFORE any calculation instruction.	want your best prediction — use what you already know from Lessons 1 to 3. Write one sentence for each scenario. You have 3 minutes — work alone.' Set a visible timer. After 3 minutes, cold-call 4 learners (two per scenario): 'Tell us your prediction and why.' Allow WAIT TIME of 12 seconds after each cold-call before probing. Ask: 'Does direction change your prediction? How?' Do NOT confirm or correct yet — record predictions on the board under the heading 'Our Predictions' so they can be revisited in Phase 3.	schema about direction and sign from Lessons 1–3, creating productive cognitive dissonance when their predictions are tested against calculated results. This directly echoes the Timer Race puzzle: learners who predicted direction 'doesn't matter' are set up for the same surprise the zigzag walker created.	(e.g., 'I think velocity could be negative because he turned back'). Teacher scans 6–8 books during the 3-minute window to gauge whether learners are defaulting to speed (scalar) thinking. If the majority ignore direction, note this for targeted questioning in Phase 3.
Observe Phase 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4–5 receive one tiered problem card each. Four problem sets are used: (1) KIPCHOGE PROBLEM — Eliud Kipchoge runs 500 m east in 90 s, then turns and runs 200 m west in 40 s. Calculate: (a) total distance, (b) total displacement, (c) average speed, (d) average velocity for the whole journey. (2) MATATU PROBLEM — A matatu on Thika Road increases velocity from 20 m/s to 35 m/s northward in 10 s. Calculate acceleration. Then the matatu brakes from 35 m/s to 0 m/s in 7 s. Calculate deceleration. What does the sign tell you? (3) MANGO PROBLEM — A mango falls from rest from a tree 7.2 m tall near Lake Victoria. It reaches the ground with velocity 12 m/s downward. Calculate (a) time of fall, (b) acceleration. Compare to g = 10 m/s². (4) BODA-BODA PROBLEM — A boda-boda on Nakuru highway travels at 15 m/s east. It overtakes a vehicle and 8 s later is moving at 27 m/s east. Calculate acceleration. If it then turns west and reaches 10 m/s	Distribute problem cards. Say: 'Each group has a different real Kenyan motion situation. Your job is to solve it completely — show every step — and be ready to explain to the class why direction appears in your answer. You have 20 minutes.' Circulate every 3–4 minutes. At each group, crouch to learner level and ask one targeted question without giving answers: Kipchoge group — 'You have two different time intervals — how will you handle that for average velocity?'; Matatu group — 'If deceleration gives you a negative number, what does that mean physically?'; Mango group — 'What is the initial velocity of a falling object released from rest?'; Boda-boda group — 'When the boda-boda moves west, how should you write that velocity compared to the eastward velocity?' If a group finishes early, pose extension: 'What would uniform acceleration look like on a velocity–time graph? Sketch it.' At 18 minutes, give a 2-minute warning: 'Finalise your board work	Structured Group Problem Solving with Contextual Anchoring — assigning each group a distinct Kenyan scenario prevents answer-sharing while ensuring all four contexts are explored by the class collectively. The requirement to connect their answer to the Timer Race phenomenon keeps the anchoring phenomenon alive and prevents the mathematics from becoming decontextualised symbol manipulation.	Observable indicators during circulation: (a) Learners correctly identify u and v (initial and final velocity) before substituting into a = (v−u)/t; (b) answers include units AND a direction word or sign; (c) at least one group member can explain verbally why the sign matters. Red-flag indicator: group writes acceleration as a positive number for the braking matatu — teacher intervenes with: 'The matatu is slowing down — what should happen to velocity over time? So is v larger or smaller than u here?'

	west in 5 s from rest, calculate that acceleration and discuss the sign. Groups complete all working in exercise books and prepare a 3-minute board presentation showing: formula used, substitution, answer with units and direction, and one sentence connecting their answer to the Timer Race phenomenon.	— one clear worked solution visible to the whole class.'		
Explain Phase 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each group presents their problem and solution to the class (3 minutes per group, 12 minutes total). During presentations, the rest of the class acts as a 'Mathematics Panel': they listen, then ask one clarifying question or offer one respectful correction. After all presentations, the teacher leads a whole-class sensemaking discussion connecting all four solutions. Learners then individually write a 'Sensemaking Summary' in their books: 'Velocity and acceleration are vectors because ______. The sign of an answer tells us ______. This connects to the Timer Race because ______.'	For each presentation, stand to the side of the room (not the front) so the presenting group owns the space. After each group finishes, use cold-call: select 2 learners from the audience to respond — one to ask a question, one to agree or challenge. Allow WAIT TIME of 12 seconds before accepting responses. After all four presentations, draw a comparison table on the board with columns: Scenario | Formula Used | Answer (with sign) | What the Sign Means | Link to Timer Race Puzzle. Complete the table collaboratively by calling on different learners for each cell. Say: 'Remember the zigzag walker in the Timer Race? They had the same distance as the straight-line walker but ended in a different place. Which of today's problems shows us the same mathematical idea most clearly?' Allow WAIT TIME of 15 seconds. Cold-call 3 learners. Finally, return to the 'Our Predictions' column on the board: 'Let's check your predictions from the start of the lesson. Who predicted correctly that direction would change the answer — and why were you right or wrong?' Cold-call 4 learners (including at least 2 who had incorrect predictions).	Gallery-Style Peer Presentation with Comparison Table — presenting multiple real-world scenarios side-by-side in a structured table externalises the pattern that direction (sign) is always mathematically present in vector quantities. The explicit link back to the Timer Race phenomenon in every table row reinforces the unit's central argument: numerical values alone are insufficient to describe motion. The individual Sensemaking Summary forces each learner to articulate this argument in their own words, which research identifies as a high-yield consolidation strategy.	Observable indicators: (a) During presentations — at least 3 of 4 groups correctly include direction/sign in their final answer; (b) During the comparison table — learners can explain a negative acceleration as deceleration without being prompted; (c) In the Sensemaking Summary — learners use the words 'vector', 'direction', and 'sign' correctly. Misconception to watch: learners who write 'negative acceleration means the object is moving backwards' — intervene by asking: 'Can a car be moving forward and still be decelerating? What would that look like?'

Driving Question Board (DQB) Creation 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners revisit the class Driving Question Board. Each group writes one new sticky note (or card) to add: (a) one piece of evidence they now have toward answering the driving question, and (b) one new question their problem-solving raised. Groups place their evidence notes in the 'Evidence We Have' column and their new questions in the 'Questions We Still Have' column. The class identifies which part of the driving question ('describe, graph, and predict') today's lesson best addresses.	Hand each group two sticky notes (different colours — e.g., green for evidence, yellow for new questions). Say: 'You have 4 minutes to write your group's contribution to our Driving Question Board. Green note: what mathematical evidence can we now use to describe motion more fully than we could before Lesson 1? Yellow note: what do we still NOT know that we would need to fully answer the driving question?' After groups post their notes, read 2–3 aloud and ask: 'Our driving question asks us to describe, GRAPH, and predict motion. Today we practised describing and predicting — what have we NOT done yet?' Cold-call 2 learners. Confirm that graphing is the focus of the next lessons. Say: 'Keep that yellow note in mind — in Lesson 5 we will see whether a graph can show direction and acceleration better than a formula alone.'	Driving Question Board Curation — the two-colour system (evidence vs. new questions) makes visible the epistemic progress of the class while honestly representing remaining gaps. Identifying which sub-component of the driving question ('graph') has not yet been addressed creates a motivating 'knowledge gap' that pulls learners forward into Lesson 5.	Observable indicator: Evidence notes specifically reference vector quantities (velocity, acceleration) and include direction language. New-question notes reference graphing, prediction, or real-world application — indicating learners are tracking the full scope of the driving question. A group whose evidence note only says 'we learned formulas' receives a probing question: 'What do the formulas tell us about direction that speed and distance could not?'
Model Building Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Individually, learners update their personal 'Motion Model' — a diagram they have been building since Lesson 1 to represent how objects move in a straight line. Today they must add: (1) the velocity formula with labels showing it is a vector (arrow notation); (2) the acceleration formula with labels for u, v, t; (3) at least one annotated example from today's problems showing positive and negative values with physical meaning; (4) a written statement: 'Direction matters in my model because ______.' Learners share their updated model with a partner and give one specific piece of feedback using the stem: 'Your model shows ______ clearly, but I	Say: 'Since Lesson 1 you have been building a personal model of linear motion. Today's lesson adds two new mathematical tools to that model. Take 7 minutes to update your diagram — make sure someone who missed today's lesson could look at your model and understand what velocity and acceleration mean, how to calculate them, and why direction is encoded in the sign.' Circulate and look specifically for: arrow notation on velocity, correct formula layout for acceleration, and the direction statement. After 7 minutes, say: 'Share your model with your partner. Use the feedback stem on the board. You have 3 minutes.' Cold-call 2 pairs	Cumulative Model Revision with Peer Feedback — requiring learners to physically update a running diagram (rather than starting fresh each lesson) makes the accumulation of mathematical understanding visible and personalised. The peer feedback protocol (specific stem) develops mathematical communication skills and surfaces residual misconceptions before the graphing lessons begin. The closing prompt bridges to Lesson 5 by framing the graph as a new representational tool for the same physical reality.	Observable indicators: (a) Model includes both $v = s/t$ and $a = (v-u)/t$ with correct labelling; (b) at least one worked example from today is annotated with direction/sign; (c) the direction statement is specific and connected to the Timer Race or today's problems — not generic. Exit ticket (collected at the door): 'A bus in Mombasa accelerates from rest to 18 m/s northward in 6 seconds. Write the acceleration with correct sign, units, and direction. In one sentence, explain what it would mean if the answer were negative.' Correct response: $a = (18-0)/6 = +3\ m/s^2$ northward; a negative value would mean the bus is decelerating (slowing down)

	notice ______ is missing or unclear.'	to share one piece of feedback they gave or received. Close by saying: 'Your model is growing. In Lesson 5 we will add a completely new layer — instead of formulas, we will use graphs. Ask yourself tonight: can a graph show direction as clearly as a + or − sign does?'		or accelerating southward.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did all four groups correctly encode direction (sign) in their final velocity and acceleration answers, or did some groups strip the sign and treat the answer as a scalar? If so, which contexts (Kipchoge, matatu, mango, boda-boda) produced the most scalar thinking — and why might that context have obscured the vector nature? (2) During peer presentations, were learners in the audience genuinely engaging as a 'Mathematics Panel', or were they passive? What one structural change (e.g., written panel question before the presentation starts) might increase audience accountability in the next lesson? (3) Review the exit tickets: what proportion of learners correctly included both units and direction in the Mombasa bus problem? If fewer than 70% succeeded, consider opening Lesson 5 with a 5-minute revisit of sign conventions before introducing velocity–time graphs. (4) Did the Sensemaking Summaries show genuine connection to the Timer Race phenomenon, or were they generic formula restatements? Use 3–4 exemplary summaries (anonymised) as mentor texts at the start of Lesson 5 to raise the quality of explanatory writing across the class.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that four different Kenyan motion situations — Eliud Kipchoge's marathon run, a matatu braking on Thika Road, a mango falling near Lake Victoria, and a boda-boda overtaking on Nakuru highway — all required us to use direction (positive or negative signs) to fully describe velocity and acceleration. We noticed that the same numerical value of acceleration could mean speeding up or slowing down depending on its sign.
What did I learn?	Velocity = displacement ÷ time and is a vector — its sign tells us direction. Acceleration = (v − u) ÷ t and is also a vector — positive acceleration means speeding up in the chosen direction, negative acceleration (deceleration) means slowing down, and zero acceleration means constant velocity. These two formulae are more powerful than speed and distance alone because they encode direction mathematically.
How does this explain the phenomenon?	We can now explain why the Timer Race puzzle is not just a curiosity — it is a mathematical necessity. When the zigzag walker retraced steps and ended in a different place despite covering the same distance, the missing information was direction. Today's problems showed us that velocity and acceleration carry direction inside their sign. Without the sign, we cannot predict where Kipchoge ends up, whether the matatu will stop in time, or how long the mango takes to fall. Direction is not optional — it is built into the mathematics of motion.

LESSON 5 (80 minutes): Drawing Displacement–Time Graphs: Reading Motion from a Line	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will draw accurate displacement–time graphs from data tables and interpret the features of those graphs to distinguish between stationary, uniform, and non-uniform motion.
Knowledge	Learners know that displacement is a vector quantity with both magnitude and direction; that velocity is the rate of change of displacement with time (v = displacement ÷ time); and that uniform velocity produces a straight-line displacement–time graph while non-uniform velocity produces a curve.
Skills	Learners can: (1) transfer data from a table into a correctly labelled displacement–time graph on grid paper; (2) calculate the gradient of a displacement–time graph to determine velocity; (3) identify segments of a graph that represent stationary, uniform, and non-uniform motion; and (4) interpret positive and negative gradients as motion in opposite directions.
Attitudes	Learners appreciate the power of graphical representation as a precise mathematical language for communicating motion, and value accuracy and careful labelling as hallmarks of scientific integrity.
Key Inquiry Question	How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Purpose in Storyline	In Lessons 1–4 learners experienced the Timer Race, defined displacement vs. distance, and calculated scalar and vector quantities numerically. In Lesson 5 they now translate those numbers into visual graphs — a crucial step toward answering the driving question. Using the data tables generated during the timer-race activity and the SGR Nairobi–Mombasa supporting phenomenon, learners discover that every feature of a displacement–time graph (slope, flatness, curve, negative gradient) tells a precise story about motion. This sets the foundation for Lesson 6, where gradients will be formally linked to velocity and acceleration graphs.
Safety Notes	No hazardous materials are used. Learners use rulers and pencils on grid paper — remind learners to pass rulers handle-first and to keep workspaces clear of bags to prevent trips.

B. LESSON OVERVIEW
Learners begin by revisiting the Driving Question Board and recalling the Timer Race puzzle. They predict what a graph of displacement against time would look like for each of the three timer-race walkers. They then receive printed data tables (one from the timer-race activity, one modelling an SGR train journey between Nairobi and Mombasa) and practise drawing accurate displacement–time graphs on grid paper in pairs. Through guided sensemaking, they connect graph features — gradient, flat sections, negative slope — to physical meaning. The lesson closes with learners updating their cumulative motion model and posting new evidence claims on the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:**	Learners individually sketch (without data) what they think a displacement–time graph would	Open with: 'Before we plot a single data point today, I want your brain's best guess on paper.'	Predict–Sketch–Share: Asking learners to commit a sketch to paper before instruction activates	Circulate during the 4-minute silent sketch. Observable indicators of readiness: (1) learner correctly

[Net force](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	look like for each of the three timer-race walkers: (A) the straight-line walker moving at steady pace, (B) the zigzag walker moving at varying pace, and (C) the walker who retraced steps back to a midpoint. They annotate their sketches with one sentence explaining their reasoning. Learners then share predictions with a partner using a Think-Pair-Share protocol before three volunteers share with the class.	Display the three walker descriptions from the Timer Race on the board alongside a blank displacement–time axis. Instruct: 'Individually, sketch what you think each walker's graph looks like. You have 4 minutes — no talking yet.' [Allow 4 minutes silent work.] Then: 'Turn and share your sketch with your partner. Explain WHY you drew it that way — not just what it looks like.' [Allow 3 minutes.] Cold-call 3 learners (aim for one strong, one partial, one unexpected sketch): 'Achieng, show us your sketch for Walker C and tell us what the negative part of your graph means to you.' [WAIT TIME: 12 seconds after question before accepting answers.] Record two or three contrasting predictions on the board under the heading 'Our Predictions' — do NOT correct them yet. Say: 'Hold onto these — by the end of today you will know exactly which of these is right, and why.'	prior knowledge from Lessons 1–4, surfaces misconceptions (e.g., learners who draw 'zigzag' graphs for the zigzag walker, confusing the path shape with the graph shape), and creates productive cognitive conflict that motivates the Observe phase.	places displacement (not distance) on the y-axis; (2) learner draws a flat horizontal line for a stationary period; (3) learner attempts a negative slope for Walker C's return. Flag learners who label the y-axis 'distance' — these learners need targeted re-voicing during the Explain phase.
Observe Phase 📹 **VIDEO:** [Net force](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Pairs receive two printed data tables. Data Table 1 is drawn from the Timer Race: it records Walker A's displacement (in metres, from the starting point, with direction defined as positive toward the finish line) at 10-second intervals over 60 seconds, including a return segment. Data Table 2 models an SGR train departing Nairobi (displacement = 0 km) toward Mombasa, recorded at 30-minute intervals over 4.5 hours, with a scheduled stop at Mtito Andei. Pairs plot both graphs on pre-printed grid paper (axes already labelled with scale). They use a ruler for straight segments and a smooth curve for non-	Distribute materials and say: 'You have two data tables and two grids. Your job is to be a precise graph-drawing machine — every point plotted, every axis labelled with units, every segment joined correctly.' [Model the first two data points of Table 1 on the board demonstrating correct plotting technique — cross marks, not dots.] 'Notice: I am using a cross, not a dot, so the exact point is clear.' Circulate continuously. At the 8-minute mark, pause the class: 'Hands up — who has found a segment where displacement is NOT changing? What does that tell you about the walker?' [WAIT TIME: 10 seconds.] Cold-call 2	Data-to-Graph Translation with Structured Annotation: Physically plotting data points and annotating features (flat segments, steep segments, curves) forces learners to connect numerical values to visual patterns. Circling flat segments and hypothesising about their meaning scaffolds the leap from 'what I see' to 'what it means physically' — directly linking back to the Timer Race puzzle (Walker C's return is visible as a descending line, resolving earlier cognitive conflict).	Collect one graph per pair at the end of this phase for a 2-minute gallery scan. Observable indicators of proficiency: (1) axes correctly labelled with quantity and unit (displacement/m; time/s); (2) scale consistent and appropriate; (3) flat segments correctly identified; (4) points joined with a ruler where data is linear. Common error to watch: learners drawing a 'zigzag' line to mimic the physical path — address whole-class if observed in more than 3 pairs.

	uniform segments. Each pair must: (1) plot all points accurately, (2) join points appropriately (straight line or curve), (3) circle any segment that looks 'flat' and write a hypothesis about what it means, and (4) mark the steepest segment with an arrow.	learners. At the 15-minute mark: 'Now look at your SGR graph — find the Mtito Andei stop and circle it. Describe what the graph looks like there to your partner.' [WAIT TIME: 12 seconds.] Close the Observe phase by asking: 'What is the steepest segment on your SGR graph telling you about the train's motion?' Cold-call 2 learners before transitioning.		
Explain Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class engages in a structured whole-class discussion to interpret the graph features. Learners use their completed graphs and the vocabulary wall (displacement, velocity, gradient, stationary, uniform, non-uniform) to explain: (1) what a flat horizontal segment means; (2) what a steep positive slope means versus a gentle positive slope; (3) what a negative slope means in the context of the Timer Race and the SGR; and (4) how to calculate velocity from the graph using gradient = rise ÷ run = Δdisplacement ÷ Δtime. Learners then individually calculate the velocity of Walker A during two different segments and the velocity of the SGR train between Nairobi and Mtito Andei.	Begin with a reference back to predictions: 'Let's return to our predictions on the board. Raise your hand if your sketch for Walker C looked anything like what your graph shows now.' [Pause — acknowledge hands without judgement.] Display a large annotated exemplar graph on the board. Use a 'gradient hunt': 'Pick the steepest segment on your Walker A graph. Using rise over run — Δdisplacement divided by Δtime — calculate the gradient. That gradient IS the velocity. Work it out now — 3 minutes, silent, individual.' [WAIT TIME: Allow 3 full minutes.] Cold-call 3 learners for their answers: 'Kamau, what gradient did you get for segment 1? What does that number mean physically — in words, not just numbers.' [WAIT TIME: 12 seconds.] Address the concept of negative gradient directly: 'When Wanjiru retraced her steps in the Timer Race, her displacement decreased. What does the graph look like? What sign is the velocity?' Build the formal statement on the board: 'Gradient of a displacement–time graph = velocity. Positive gradient = moving away from reference point. Zero gradient = stationary. Negative gradient = moving back	Gradient Hunt with Formal Generalisation: Learners first calculate gradients for specific segments (concrete), then the teacher formalises the pattern into a transferable rule (abstract). Connecting the negative gradient explicitly to Walker C's return journey and the matatu example from the driving question anchors abstract algebra in lived Kenyan experience and resolves the original cognitive conflict from Lesson 1.	Observe individual gradient calculations. Proficiency indicators: (1) learner correctly selects two points on a straight segment (not random points); (2) correctly computes Δdisplacement and Δtime with correct units; (3) states velocity with a sign and a unit (e.g., '+2.5 m/s' not just '2.5'); (4) can verbally interpret the sign. Misconception to probe: learners who compute gradient using distance instead of displacement — ask: 'Did you use the y-axis value or the actual path length?'

		toward or past the reference point.' Connect to the driving question: 'So can a displacement–time graph tell us direction? Yes — the sign of the gradient tells us direction. This is exactly why the speedometer of a matatu cannot tell you which direction Nairobi is, but a displacement–time graph can.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one new 'evidence claim' on a sticky note in the format: 'I now have evidence that… [claim] … because… [graph feature].' They post it under the relevant driving question on the class Driving Question Board. Learners also write one remaining question — something today's lesson raised but did not yet answer — on a different coloured sticky note. The class does a 2-minute silent gallery walk to read peers' claims before returning to seats.	Distribute two sticky notes per learner (different colours). Say: 'In the evidence colour, write one thing you now have graph evidence for — use the sentence stem: I now have evidence that… because my graph shows… In the question colour, write one thing you are still wondering.' [Allow 4 minutes.] 'Now post your evidence note under the driving question it helps answer, and your question note in the open questions section.' [Allow 2 minutes for posting.] Facilitate a 2-minute silent gallery walk: 'Walk around, read three claims that are NOT yours, and put a tick on any claim you agree with.' After gallery walk: 'I noticed several of you wrote questions about what happens when velocity itself is changing — that is exactly where we are going in Lesson 6.' [Preview hook.] Cold-call 2 learners: 'Fatuma, read out your evidence claim — how does it connect to the Timer Race puzzle we started with?'	Evidence Claim Posting with Gallery Walk: The structured sticky-note protocol requires learners to articulate the logical link between their graph observation (evidence) and their conceptual claim — a core scientific practice. The gallery walk creates peer accountability and helps the teacher rapidly audit the range of understanding across the class without collecting books.	Read sticky notes as they are posted. Proficiency indicator: evidence claim includes a specific graph feature (e.g., 'the flat segment,' 'the negative gradient') linked to a physical meaning (e.g., 'the train was stationary,' 'Walker C was moving backward'). Flag vague claims (e.g., 'I learned about graphs') for individual follow-up at lesson start in Lesson 6.
Model Building Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners return to their cumulative motion model — a diagram they have been building since Lesson 1 representing the Timer Race scenario mathematically. In this lesson they add a new layer: a displacement–time graph sketch for each of the three walkers,	Say: 'Take out your cumulative motion model. We are going to add the graphical language layer today — this is the most powerful tool we have added so far.' Display the model-building prompt on the board: '(1) Draw a displacement–time graph sketch for each of the	Cumulative Model Revision: Adding a graph layer to an existing model makes conceptual growth visible to learners. Requiring annotation of segment types and velocity signs on the same diagram that already contains the numerical calculations from earlier	Circulate and review updated models. Proficiency indicators: (1) all three walkers' graphs are present; (2) Walker C's graph includes a clearly visible negative-gradient segment; (3) the formula gradient = $\Delta s/\Delta t$ is correctly written and linked to the graph; (4) learner

for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	annotated with labels identifying stationary segments, uniform motion segments, non-uniform segments, and the sign of the velocity in each segment. They also add the formal relationship: gradient = velocity = Δs/Δt. Pairs compare their updated models and agree on a 'best version' to display.	three Timer Race walkers. (2) Label: stationary / uniform / non-uniform sections. (3) Mark the sign of velocity on each segment. (4) Write the formula: gradient = Δs ÷ Δt = velocity.' [Allow 8 minutes for individual additions.] Then: 'Compare with your pair — do your graphs for Walker C agree on which segments are negative? Discuss for 2 minutes.' [WAIT TIME: Allow 2 full minutes.] Cold-call 2 pairs: 'Show me Walker C's graph on your model and explain one segment to the class.' Close with a connection to the driving question: 'Every time we add a new layer to this model, we get closer to fully describing motion. Today we added graphical language. What do you think we are still missing to fully predict motion?' [WAIT TIME: 12 seconds.] Accept 2–3 responses — record key words (likely: acceleration, changing velocity) on the board as a preview of Lesson 6.	lessons reinforces the coherence between algebraic and graphical representations — a key mathematical practice in the CBE framework.	can verbally explain at least one segment when cold-called. Learners whose models omit direction information (negative slope) should be flagged for small-group consolidation at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Predict phase, how many learners confused the shape of the physical path (zigzag) with the shape of the graph? What does this tell you about residual distance–displacement confusion from Lessons 1–4 that may need revisiting? (2) Were learners able to compute the gradient correctly and attach a sign and a unit to their answer? If more than a third of the class omitted the sign or the unit, plan a 5-minute targeted starter for Lesson 6 using a simple Walker A segment before moving to acceleration graphs. (3) Review the sticky-note evidence claims: How many learners connected the negative gradient explicitly to direction (a vector concept) rather than just saying 'going back'? This distinction is critical for Lesson 6's treatment of velocity and acceleration as vectors. (4) Were the two data tables (Timer Race and SGR) at an appropriate challenge level — did pairs finish plotting within the allocated 20 minutes? If not, consider pre-plotting the SGR table as a homework task before Lesson 6 to protect class time for interpretation. (5) Did the cumulative model updates show growing sophistication compared to Lesson 4? Photograph two or three models to open Lesson 6 with peer exemplars.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners plotted displacement–time graphs from data tables based on the Timer Race walkers and the SGR Nairobi–Mombasa journey, identifying flat segments (stationary motion), steep segments (fast uniform motion), gentle segments (slow uniform

	motion), and negative-gradient segments (motion in the opposite direction, as shown by Walker C retracing her steps).
What did I learn?	The gradient of a displacement–time graph equals the velocity of the object (gradient = $\Delta s \div \Delta t$). A positive gradient means motion away from the reference point; a zero gradient means the object is stationary; a negative gradient means motion back toward or past the reference point. This is why a displacement–time graph — unlike a speedometer or a distance value alone — can communicate both speed and direction.
How does this explain the phenomenon?	Every segment of a displacement–time graph encodes information about velocity: its steepness gives the magnitude and its sign gives the direction. This is the mathematical language the Timer Race puzzle demanded — it explains why Walker C, despite covering the 'same distance,' ends up in a completely different location, and why Eliud Kipchoge's 42 km marathon displacement is far smaller than his total distance run. Direction always matters, and graphs make direction visible.

LESSON 6 (80 minutes): Interpreting Displacement–Time Graphs	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To interpret displacement–time graphs by reading values, calculating gradients, and writing narrative journey descriptions — deepening understanding of how direction and velocity are encoded in graphical representations of motion.
Knowledge	Learners will know that: (1) the gradient of a displacement–time graph equals velocity; (2) a positive gradient indicates motion in a positive direction, a negative gradient indicates return motion, and a zero gradient (horizontal line) indicates the object is stationary; (3) velocity is a vector quantity with both magnitude and direction; (4) uniform velocity appears as a straight line on a displacement–time graph, while non-uniform velocity appears as a curve.
Skills	Learners will be able to: (1) read displacement and time values from a displacement–time graph; (2) calculate the gradient of a straight-line segment to determine velocity; (3) identify stationary periods, constant-velocity periods, and changing-velocity periods from a graph; (4) write an accurate narrative description of a journey from its displacement–time graph.
Attitudes	Learners will appreciate that graphical representations are powerful mathematical tools that reveal information (especially direction) that raw numerical values alone cannot communicate — connecting back to why Eliud Kipchoge's 42 km marathon displacement is near zero.
Key Inquiry Question	How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Purpose in Storyline	In Lessons 1–5, learners established the distinction between distance and displacement, speed and velocity, and began calculating velocity and acceleration numerically. In Lesson 6, learners now encounter the graphical language of motion: given pre-drawn displacement–time graphs of a boda-boda journey in Kisumu and a jogger in Karura Forest, Nairobi, they must decode the story hidden in each graph — reading values, computing gradients as velocities, and identifying direction changes. This is a pivotal evidence-gathering lesson: learners discover that the slope (gradient) of a displacement–time graph IS velocity, and that a negative slope means the object is moving back toward the origin, making direction visible on a graph in a way a speedometer never could.
Safety Notes	No physical hazards in this lesson. Learners use rulers, pencils, and printed graph sheets. Ensure graph worksheets are clearly printed with legible axes. If digital devices are used for graph simulations, apply standard screen-use guidelines. Remind learners to handle rulers safely.

B. LESSON OVERVIEW

Learners receive two pre-drawn displacement–time graphs: (1) a boda-boda rider making a delivery run in Kisumu — travelling from the ferry terminal toward Kondele market, stopping, and returning partway; and (2) a jogger completing an out-and-back run in Nairobi's Karura Forest. Working in pairs, learners systematically read values from each graph, calculate gradients of individual segments to determine velocity, identify stationary periods and direction changes, and compose written narrative descriptions of each journey. A whole-class gallery discussion connects gradient calculations back to the unit's driving question, cementing the insight that the sign of the gradient encodes direction — the very information a speedometer (or raw distance figure) cannot provide. The lesson closes with learners updating the Driving Question Board and revising their cumulative motion models to include graphical representation conventions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two unlabelled, blank-axis displacement–time graph sketches on the board (Sketch A: a straight line rising steeply, then a horizontal segment, then a straight line falling back to zero; Sketch B: a gentle upward curve). Without any axis labels or numbers, learners individually write in their notebooks: (1) What do you think is happening to the object in Sketch A? Describe the journey in words. (2) Which sketch do you think shows a faster object — and how can you tell just from the shape? (3) What does a horizontal (flat) section of any motion graph tell you? Learners then share predictions with a partner (Think-Pair) before three cold-calls are taken by the teacher.	Display the two sketches on the board or projected screen. Say: 'Before we put any numbers on these graphs, I want you to be a detective. Look only at the shape of each line. In your notebook, answer these three questions — you have 3 minutes, work silently and independently.' [Set 3-minute timer.] Circulate and scan notebooks; note two or three interesting or contrasting predictions to highlight later. After 3 minutes, say: 'Turn to your partner — share what you wrote for question 1. You have 90 seconds.' After pair-share, cold-call THREE learners: 'Amina, what journey did you imagine for Sketch A?', 'Brian, do you agree with Amina — what would you add?', 'Zawadi, which sketch shows the faster object and why?' After each response, provide 10–15 seconds WAIT TIME before affirming or gently probing: 'Interesting — what in the shape made you think that?' Do NOT confirm or correct predictions yet. Close by saying: 'Hold onto those ideas — the graphs we analyse today will tell us if your detective instincts were right.'	Strategy: Anticipation Sketching. By interpreting unlabelled graph shapes before receiving data, learners activate prior knowledge of gradient and rate of change from earlier mathematics strands, and surface any conflation of 'steeper = more distance' versus 'steeper = faster'. This productive struggle primes learners to notice gradient as the key feature during the Observe phase.	Observable indicator: Learners' written predictions use directional language (e.g., 'going forward then coming back', 'stopped', 'moving faster'). Teacher scans for learners who describe the horizontal segment as 'going slowly' rather than 'stationary' — these learners need targeted support in the Observe phase. Teacher notes at least 3 learners' predictions to revisit during the Explain phase closure.
Observe Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data	Learners receive the printed Graph Worksheet containing two fully labelled displacement–time graphs. GRAPH 1 — The Kisumu Boda-Boda: A boda-boda rider leaves the Kisumu ferry terminal (displacement = 0 m) and travels toward Kondele market. The graph shows: Segment A (0–4 min): displacement increases linearly from 0 to 1200 m; Segment B (4–6 min): displacement remains	Distribute the printed Graph Worksheets face-down. Say: 'When I say turn over, you will see two real journeys — one by a boda-boda rider in Kisumu, one by a jogger in Karura Forest. Your job is to read the graph carefully — like reading a map — and fill in the observation table for each segment. Work in pairs. You have 18 minutes.' [Flip timer visible to class.] Walk the room	Strategy: Structured Data Table with Directed Graph Reading. The segmented observation table scaffolds learners to treat each line segment as a discrete data event, isolating gradient calculation from narrative interpretation. By labelling Δs as signed (positive or negative), the table forces learners to confront direction as a mathematical quantity — not just a verbal description — directly	Observable indicators: (1) Learners correctly record negative Δs values for return segments (Segment C of Graph 1; Segment R of Graph 2). (2) Learners calculate gradients with correct units (m/min or m/s). (3) Learners correctly identify Segment B gradient = 0 and link it to 'stationary'. Teacher listens for pairs who calculate gradient as Δt/Δs (inverted) — these pairs

Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	constant at 1200 m; Segment C (6–10 min): displacement decreases linearly from 1200 m back to 400 m. GRAPH 2 — The Karura Forest Jogger: A jogger starts at the Karura Forest main gate (displacement = 0 m). Segment P (0–5 min): displacement increases linearly from 0 to 800 m; Segment Q (5–8 min): displacement increases linearly from 800 m to 1400 m (shallower gradient); Segment R (8–13 min): displacement decreases linearly from 1400 m to 0 m. Working in pairs, learners complete a structured observation table for each segment: (a) Start displacement, (b) End displacement, (c) Time taken, (d) Change in displacement (Δs), (e) Gradient = Δs/Δt, (f) What does the gradient tell you about the motion? Learners also mark with a coloured pen where direction changes occur on each graph.	systematically — spend no more than 90 seconds per pair. Use targeted prompts, NOT answers: 'How did you calculate the gradient for Segment A?', 'What does a negative Δs mean for the boda-boda rider — which direction is he going?', 'You wrote 300 m/min for Segment A — can you convert that to metres per second?' After 10 minutes, pause the class with a hand signal: 'Freeze — quick check. Everyone look at Segment B of Graph 1. What number did you get for the gradient? Show me on your fingers: zero, positive, or negative.' [Scan responses — address any zeros-vs-horizontal confusion immediately.] After 18 minutes total, say: 'Finish your last entry and put your pens down. We will share findings now.'	connecting to the unit's core vector theme.	need a targeted prompt: 'Gradient is rise over run — which axis is the rise here?' Collect 3–4 worksheets at random at the end of this phase for rapid review before the Explain phase.
Explain Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes for a structured whole-class sensemaking discussion. The teacher projects Graph 1 (Kisumu Boda-Boda) on the board and calls pairs to share their gradient calculations segment by segment, building a class consensus table on the board. Learners then tackle the core conceptual question: 'What IS the gradient of a displacement–time graph?' Learners articulate — in their own words — that gradient = velocity, and that the sign of the gradient tells you the direction of travel. They then apply this to write a full narrative description of the boda-boda journey in 4–5 sentences. Next, learners independently (no pair support) write the narrative for	Project Graph 1. Say: 'Let's build the class answer together. Pair 1 — give me the gradient for Segment A and tell me what it means.' [Cold-call Pair 1. Allow 10 seconds WAIT TIME after their answer.] 'Does anyone disagree or want to add a unit?' [Cold-call a second pair.] Record the agreed gradient on the board with units. Repeat for Segments B and C. Then ask the key conceptual question: 'So — in one sentence — what is the gradient of a displacement–time graph?' [10 seconds WAIT TIME. Cold-call three different learners — ensure at least one learner who was silent in Phase 2 is called.] Write the class-constructed definition on the board: GRADIENT OF	Strategy: Co-construction of a Consensus Table + Claim–Evidence–Reasoning (CER) Narrative Writing. Building the gradient table collectively ensures all learners have correct values before attempting narrative writing. The CER narrative task requires learners to translate mathematical outputs (gradient values, signs) into scientific claims about motion — this dual representation (algebraic ↔ narrative) deepens conceptual understanding and directly mirrors the NGSS practice of constructing explanations. The return to the anchoring phenomenon (drawing the race graphs) creates a bridging moment that consolidates the lesson's key insight.	Observable indicators: (1) Learners' narrative descriptions include direction (e.g., 'moving back toward the ferry terminal') not just speed. (2) Learners' narratives correctly identify the stationary period. (3) During the phenomenon sketch task, learners draw the zigzag walker's graph with a lower gradient (smaller displacement per unit time) than the straight-line walker's graph — indicating they understand displacement, not distance, determines gradient. Teacher collects 5–6 narrative paragraphs for written feedback before Lesson 7.

	the Karura Forest jogger's journey. The class closes by returning to the Timer Race Puzzle: 'If we had drawn a displacement–time graph for the zigzag walker and the straight-line walker in our opening race, what would the graphs look like? Would their gradients be the same?' Learners sketch their answers on mini whiteboards or in notebooks.	DISPLACEMENT–TIME GRAPH = VELOCITY. Circle the word VELOCITY. Say: 'Why velocity and not speed?' [WAIT TIME 12 seconds. Cold-call.] Affirm the vector/direction answer. Now say: 'In your notebook, write a 4–5 sentence story of the boda-boda rider's journey using the numbers from the graph. Include direction, velocity, and what happens at each segment. You have 5 minutes — this is individual silent work.' After 5 minutes, share two contrasting narratives (one strong, one with directional language missing) under a visualiser or by reading aloud — ask the class: 'What important information is missing from the second story?' Transition: 'Now do the same independently for the Karura Forest jogger.' Finally, pose the phenomenon question and give 3 minutes for the mini-whiteboard sketch before cold-calling three pairs to share.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. Working in their pairs, they write two sticky notes: (1) a GREEN 'Evidence' note — one thing they now know for certain about how graphs describe motion (e.g., 'The gradient of a displacement–time graph = velocity, including direction'); (2) an ORANGE 'New Question' note — a question this lesson has raised for them (e.g., 'What does a curved displacement–time graph mean — is the velocity changing?', 'Can we draw a velocity–time graph from this data?'). Pairs post their notes in the appropriate columns of the DQB. The teacher reads three or four new questions aloud and the	Direct learners to the DQB. Say: 'You each have two sticky notes. Green = something you can now prove with evidence from today's graphs. Orange = a new question today's lesson opened up. You have 3 minutes — write clearly, one idea per note.' After 3 minutes, invite pairs to post their notes. Read four orange 'question' notes aloud — for each one, ask: 'Thumbs up if this is a question YOU also have.' Tally the top two questions. Say: 'Look at these new questions — especially this one about curved graphs. That is exactly where we are heading in Lesson 7. You are doing what real mathematicians do: using evidence to ask better questions.'	Strategy: Driving Question Board — Evidence and Question Sorting. The DQB makes the knowledge-building arc of the unit visible. By explicitly distinguishing 'evidence we now have' from 'questions we still need to answer', learners practise the NGSS practice of obtaining, evaluating, and communicating information, and experience science/mathematics as a cumulative inquiry rather than a set of isolated lessons.	Observable indicator: Green evidence notes correctly reference gradient = velocity and sign = direction. Orange question notes reveal forward-looking curiosity about velocity–time graphs or curved graphs — indicating learners are extrapolating beyond the lesson content, a sign of deep engagement. Teacher photographs the DQB at the end of each lesson to track the evolution of class understanding across the unit.

	class votes (thumbs up) on which questions they most want answered in upcoming lessons. The teacher adds a new DQB anchor question: 'How can we draw AND interpret a velocity–time graph — and what new information would it give us?'	Point to the unit driving question displayed prominently: 'How does today's work help us answer — why does direction always matter?' Cold-call two learners to respond in one sentence each.		
Model Building Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative Motion Model — a visual/diagrammatic representation they have been building since Lesson 1 (which began as a simple sketch of the Timer Race Puzzle and has grown to include annotated diagrams of distance vs displacement, speed vs velocity, and numerical velocity/acceleration calculations). In Lesson 6, learners add a new panel to their model: a hand-drawn displacement–time graph that represents a journey of their own invention — set in a Kenyan context of their choosing (e.g., a matatu journey from Westlands to the CBD in Nairobi, a fisherman at Lake Victoria rowing out and back, a runner at the Eldoret Athletics Training Centre). The graph must include: at least three segments (one stationary, one positive-velocity, one negative-velocity); labelled axes with units; gradient calculations annotated on the graph; and a 3-sentence narrative caption. Learners also add a 'Key Insight' box to their model: 'Gradient of a displacement–time graph = velocity. A negative gradient means the object is returning toward the origin. A horizontal line means the object is stationary.'	Say: 'Open your cumulative Motion Model to the next blank panel. You are going to add a displacement–time graph — but this time, YOU invent the journey. Set it somewhere in Kenya — make it real, make it yours. Your graph must have at least three segments: one where the object is still, one where it moves away from the start, one where it moves back. Label your axes, calculate the gradient for each segment, and write a 3-sentence story beneath it.' [Allow 12 minutes for individual model building. Circulate continuously.] Targeted prompts: 'What does the gradient of this segment work out to — and what does the negative sign tell the reader?', 'Have you labelled your axes with units?', 'Can you tell me in one sentence what your Key Insight box says?' With 2 minutes remaining, say: 'Add your Key Insight box now — write the three statements we agreed on together.' Close the lesson: 'Your model now shows the mathematical language of motion — shape, gradient, sign. Next lesson, we will add velocity–time graphs and discover what new secrets THOSE graphs reveal about acceleration.' Ask learners to leave models on their desks for the teacher to scan before Lesson 7.	Strategy: Cumulative Representational Model Building (Creative Transfer Task). By inventing their own displacement–time graph in a personally chosen Kenyan context, learners perform a creative transfer — applying the lesson's concepts to a novel situation of their own design. This is a higher-order cognitive task (Bloom's Create level) that reveals whether learners can operationalise gradient, sign, and stationary segments independently, rather than simply reproducing teacher-given examples. The model's cumulative nature also reinforces that motion concepts build on one another across the unit.	Observable indicators: (1) Learner-invented graphs include a correctly drawn horizontal segment labelled with gradient = 0. (2) Return segments slope downward (negative gradient) and learners' narrative captions use directional language. (3) Gradient calculations on the graph are annotated correctly with units of m/s or m/min. (4) Key Insight box contains all three statements. Teacher collects 6–8 models overnight to assess quality of gradient annotations and narrative captions, returning them with written feedback at the start of Lesson 7.

D. TEACHER REFLECTION
"After the lesson, ask yourself: (1) Could my learners articulate — unprompted — that gradient = velocity AND that the sign of the gradient indicates direction? If not, which phase broke down — the observation table, the class consensus build, or the narrative writing task? (2) Did the cold-calls during the Explain phase reveal any persistent misconceptions (e.g., conflating horizontal graph segments with 'slow movement' rather than 'stationary')? Plan a 5-minute 'myth-buster' opener for Lesson 7 if this misconception was widespread. (3) Were learners' invented graphs in the Model Building phase sufficiently varied — did most learners default to the same shape as the boda-boda graph, or did they show genuine creative transfer? If graphs were largely imitative, increase the creative demand in future model-building tasks. (4) Review the orange DQB question notes: do learners' new questions naturally lead toward velocity–time graphs and acceleration (Lesson 7)? If learners' questions are still focused on displacement–time, consider opening Lesson 7 with a direct comparison of the two graph types side by side. (5) Did the return to the anchoring phenomenon (drawing the race graphs) land effectively? Learners should be able to say that the zigzag walker's displacement–time graph would show a lower final displacement despite the same time — if this was unclear, revisit the phenomenon sketch at the start of Lesson 7."

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners read and analysed two pre-drawn displacement–time graphs: a boda-boda rider's journey in Kisumu (ferry terminal to Kondele market, with a stop and partial return) and a jogger's out-and-back run in Nairobi's Karura Forest. They observed that different segments of each graph had different shapes — some rising steeply, some flat, some sloping downward — and recorded displacement and time values for each segment.
What did I learn?	Learners learned that the gradient (slope) of a displacement–time graph is equal to the velocity of the object for that segment. A steep positive gradient means high velocity in the forward direction; a shallow positive gradient means low velocity forward; a zero gradient (horizontal line) means the object is stationary; and a negative gradient means the object is moving back toward the starting point. Because gradient encodes direction as well as speed, displacement–time graphs display velocity — a vector quantity — not merely speed.
How does this explain the phenomenon?	Learners explained that a speedometer (or a raw distance/speed number) cannot tell you the direction of travel, but the gradient of a displacement–time graph can — a negative gradient makes the return direction mathematically visible. Applied to the anchoring phenomenon, this explains why Eliud Kipchoge's marathon displacement–time graph would end near zero gradient despite 42 km of running distance, and why the zigzag walker in the Timer Race Puzzle would show a smaller final displacement than the straight-line walker on a graph — even if both graphs cover the same total time.

LESSON 7 (80 minutes): Drawing Velocity–Time Graphs: Gradients, Acceleration, and Kipchoge's Splits	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will draw accurate velocity–time graphs from tabulated data, interpret the gradient of a velocity–time graph as acceleration, and calculate acceleration graphically — connecting these skills to the anchoring phenomenon of the Timer Race Puzzle and Kipchoge's marathon splits.
Knowledge	1. Velocity is the rate of change of displacement with time (v = displacement ÷ time) and is a vector quantity. 2. Acceleration is the rate of change of velocity with time (a = (v − u) ÷ t). 3. The gradient of a velocity–time graph equals the acceleration of the object. 4. A horizontal line on a v–t graph represents zero acceleration (uniform velocity); a positive slope represents positive acceleration; a negative slope represents deceleration.
Skills	1. Plotting accurate velocity–time graphs from tabulated data using appropriate scales and labelled axes. 2. Calculating gradient from a v–t graph to determine acceleration. 3. Interpreting the shape of a v–t graph to describe an object's motion qualitatively and quantitatively. 4. Connecting graphical representations to real-world motion scenarios.
Attitudes	Learners appreciate the power of mathematical graphs as a precise language for describing motion, and develop persistence in constructing and interpreting graphical evidence.
Key Inquiry Question	How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Purpose in Storyline	In Lessons 1–6, learners built understanding of distance, displacement, speed, velocity, and began interpreting distance–time graphs. In Lesson 7, learners now move to velocity–time graphs — a richer tool that reveals not just how fast an object is moving, but how its velocity is changing over time (acceleration). Using Kipchoge's marathon split data and a matatu journey between Nairobi and Thika, learners discover that the gradient of a v–t graph is the missing mathematical link to acceleration, advancing their answer to the driving question: direction and rate of change together give us the full picture of motion.
Safety Notes	No physical hazards in this lesson. Learners use rulers and pencils for graph work — remind learners to handle rulers carefully. Ensure graph paper is distributed without crumpling to allow accurate plotting.

B. LESSON OVERVIEW
Learners receive two sets of tabulated velocity–time data: (1) a simplified version of Eliud Kipchoge's marathon velocity splits (km/h at key race intervals) and (2) a matatu travelling from Nairobi's GPO to Thika town with varying speeds. Working in pairs, learners plot these onto velocity–time graphs, then use the gradient formula to calculate acceleration during each phase. Through guided discussion, they connect the gradient of a v–t graph to acceleration, discover what horizontal, positive-slope, and negative-slope sections mean, and revisit the Timer Race Puzzle to explain why the matatu's speedometer reading alone cannot tell you where the matatu is headed. The lesson culminates in learners updating the class Driving Question Board and revising their cumulative motion model to include acceleration arrows.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are shown two unlabelled	Display the two graphs using the	Individual-to-pair-to-class	Observable indicator: Learners'

📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	velocity–time graphs on the board (one with a horizontal line at v = 12 m/s; one with a steadily rising line from v = 0 to v = 20 m/s over 10 seconds). Working individually for 3 minutes, they write in their notebooks: (a) What do you think is happening to the object's velocity in each graph? (b) Which graph shows a matatu cruising at constant speed on Thika Road, and which shows one accelerating from the GPO bus stage? (c) How do these graphs connect to what we discovered about Kipchoge's marathon pace — did he run at perfectly constant speed throughout?	chalkboard or printed handout. Say: 'Before we learn anything new today, I want you to use what you already know. Look at these two graphs carefully and write your predictions. You have 3 minutes — no talking yet.' [WAIT 3 minutes, circulate silently, note 2–3 interesting or incorrect predictions to address later.] After 3 minutes: 'Turn to your partner and share your predictions for 1 minute.' [WAIT 1 minute.] Cold-call 3 learners: 'Amara, tell me what you think is happening in Graph A.' [WAIT 10 seconds.] 'Fatuma, do you agree or disagree with Amara — and why?' [WAIT 10 seconds.] 'Otieno, how does your prediction connect to what Kipchoge might look like on this graph?' [WAIT 12 seconds.] Record key prediction phrases on the board under the heading 'Our Predictions — we will test these today.'	prediction (Think–Pair–Share). This activates prior knowledge of velocity and graphical representation from Lessons 5–6, surfaces misconceptions (e.g., confusing steepness with speed rather than acceleration), and creates genuine curiosity about what gradient means on a v–t graph.	written predictions use the word 'velocity' correctly and attempt to describe change over time. Listen for whether learners conflate the height of the line (speed) with the slope (acceleration). Collect 3–4 notebooks at random to scan predictions before Phase 2 begins.
Observe Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners receive Handout 7A: two data tables. Table 1 — Kipchoge's Marathon Splits (simplified): time checkpoints at 0, 10, 20, 30, 42 minutes into a training run, with corresponding velocities in m/s (e.g., 0 m/s at start, 5.5 m/s at 10 min, 5.8 m/s at 20 min, 5.6 m/s at 30 min, 5.2 m/s at 42 min). Table 2 — Nairobi GPO to Thika Matatu Journey: time in seconds (0, 10, 20, 30, 40, 50, 60) with velocity in m/s (0, 4, 8, 12, 12, 12, 6). Working in pairs, learners: (a) Plot both datasets on separate velocity–time graphs on graph paper, choosing appropriate scales, labelling axes with units, and giving each graph a title. (b) Draw a best-fit straight line or connect points as appropriate for each phase. (c) Annotate each	Distribute Handout 7A and graph paper. Say: 'You are now going to become data detectives. These two tables contain real-world velocity data — one from the greatest marathon runner on Earth, one from a matatu most of us have probably ridden. Your job is to plot these carefully and observe what the graph tells you that the table cannot.' [WAIT for settling — approximately 30 seconds.] Circulate actively. After 5 minutes, pause the class: 'Quick check — raise your hand if you have labelled both axes with units.' [Scan room.] 'If your axes are not labelled, fix that now — a graph without labels is like a map without place names.' Continue circulating. Prompt pairs who finish early: 'Look at the slope of each section.	Data-to-graph translation with annotation. By physically plotting real data and annotating sections, learners move from abstract numbers to visual patterns. Annotating 'speeding up / constant / slowing down' bridges everyday language to mathematical gradient, making the upcoming gradient calculation feel motivated rather than arbitrary.	Observable indicators: (1) Both axes are labelled with quantity and units (time in seconds/minutes, velocity in m/s). (2) Points are plotted accurately within ±0.5 small squares. (3) Annotations ('speeding up', 'constant', 'slowing down') correctly match the slope direction. Collect one graph per pair at end of phase for quick review.

	graph section with a word: 'speeding up', 'constant', or 'slowing down'. (d) Observe and record: 'What is different about the sections where the line is flat versus where it is sloping?'	Can you calculate rise ÷ run for the steepest section?' After 15 minutes, cold-call 2 pairs to share their graphs under the visualiser or by drawing on the board. Say: 'Notice what Baraka's group did with their scale — does everyone agree that scale choice changes how the graph looks but not what it means?' [WAIT 10 seconds for responses.]		
Explain Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class discussion to formalise the concept that gradient of a v–t graph = acceleration. Learners then work through three structured calculations: (1) Calculate the acceleration of the matatu during the first 30 seconds (rising section) using gradient = Δv ÷ Δt. (2) State the acceleration during seconds 30–50 (horizontal section) and justify from the graph. (3) Calculate the deceleration from second 50 to second 60. Learners write a formal definition of acceleration in their notebooks and complete two extension questions: (a) If Kipchoge's velocity decreased from 5.8 m/s to 5.2 m/s over the last 12 minutes of his training run, calculate his acceleration and explain what the negative sign means. (b) Sketch what a v–t graph would look like for a boda boda rider who accelerates, travels at constant speed past Kericho town centre, then brakes to a stop.	Begin with a direct instruction segment (8 minutes). Draw a clean v–t graph on the board with a clearly rising line. Say: 'We have a new question: what does the steepness of this line actually mean mathematically? In Lessons 5 and 6 we learned that the gradient of a distance–time graph gives us speed. Now I want you to tell me — what do you think the gradient of a VELOCITY–time graph gives us? Write one word in your notebook.' [WAIT 12 seconds.] Cold-call 4 learners. [WAIT 10 seconds each.] After responses: 'The gradient of a velocity–time graph gives us the rate of change of velocity — which we call acceleration. Let me show you the calculation.' Write on board: a = (v − u) ÷ t = gradient = rise ÷ run = Δv ÷ Δt. Work through the matatu example step by step, narrating each step aloud. Then say: 'Now you try calculations 1, 2, and 3 on your own — 8 minutes, silent work.' [WAIT 8 minutes, circulate.] After independent work, pair-share for 2 minutes. Cold-call 3 learners for answers, asking: 'What does the negative value in Kipchoge's acceleration tell us about his motion — connect it back to our Timer Race.' [WAIT 12 seconds per learner.]	Gradient as physical meaning (Concept-to-Calculation Bridge). By first asking learners to predict what gradient means, then formalising it with the formula, then applying it to Kipchoge and the matatu, the teacher builds a conceptual bridge between the visual graph feature (steepness) and the physical quantity (acceleration). This prevents rote formula use without understanding.	Observable indicators: (1) Learners correctly calculate a = (12 − 0) ÷ 30 = 0.4 m/s² for the matatu acceleration phase. (2) Learners state a = 0 m/s² for the constant velocity section with a correct graph-based justification ('the line is flat, so Δv = 0'). (3) Learners correctly interpret a negative acceleration value as deceleration. Misconception to watch: learners confusing a steep line with 'high velocity' rather than 'high acceleration' — address immediately with: 'The HEIGHT of the line tells you how fast. The SLOPE tells you how fast the speed is changing.'

Driving Question Board (DQB) Creation 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each pair receives two sticky notes. On the first (green): they write one thing they can now explain about the driving question — 'How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter?' On the second (orange): they write one new question or remaining confusion the lesson has raised. Learners post their notes on the class DQB. The class then does a 3-minute gallery walk of the new additions, and the teacher facilitates a 5-minute whole-class synthesis discussion, drawing arrows on the DQB to show how today's evidence (v–t graphs, gradient = acceleration) connects to previously posted evidence (d–t graphs, velocity formula, displacement vs. distance).	Say: 'It is time to update our Driving Question Board. Today you have gathered new mathematical evidence. On your green note, write specifically what you can now explain — use the words velocity, gradient, and acceleration in your sentence. On your orange note, write a genuine question that today's lesson raised for you — there are no wrong questions.' [WAIT 4 minutes for writing.] Direct learners to post notes on the DQB and circulate for 3 minutes reading. Reconvene: 'I see Njeri wrote — [read a strong green note]. How does this connect to the puzzle we saw on Day 1 — why Kipchoge ends up only hundreds of metres from his start after 42 km?' [WAIT 10 seconds.] Draw a line on the DQB connecting today's acceleration concept to the displacement vs. distance card from Lesson 1. Say: 'Every lesson, our mathematical language gets richer. Today we added acceleration and gradient. What do you think the next piece of the puzzle might be?' [WAIT 12 seconds, cold-call 2 learners.]	Driving Question Board synthesis (Progressive Evidence Mapping). By physically connecting new sticky notes to existing DQB cards with arrows, learners visualise how each lesson's evidence builds cumulatively toward answering the driving question. This metacognitive strategy helps learners see that science and mathematics are not isolated lessons but a growing, connected argument.	Observable indicators: (1) Green notes correctly use at least two of the three target words (velocity, gradient, acceleration) in a meaningful sentence. (2) Orange notes reveal genuine scientific or mathematical curiosity rather than off-topic questions — these surface areas needing consolidation in Lessons 8–10. (3) During gallery walk, learners can verbally connect at least one today-card to one previous-lesson card.
Model Building Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data	Learners retrieve their cumulative motion model (begun in Lesson 1 and revised each lesson). Today they add a new layer: a velocity–time graph sketch of the Timer Race Puzzle characters (the straight-line walker, the zigzag walker, and the backtracker), annotating each graph section with the calculated or estimated acceleration. They also add a key to their model: a legend showing what the gradient of a v–t graph represents, with the formula $a = \Delta v \div \Delta t$. Finally, learners write a 'model caption' of no more than	Say: 'Take out your cumulative motion model. Today we are adding the most powerful layer yet — velocity–time graphs for our three Timer Race characters. Sketch a v–t graph for each person: the straight-line walker (roughly constant velocity), the zigzag walker (varying velocity), and the backtracker (positive velocity then negative velocity). Add acceleration arrows on each graph section.' [WAIT 8 minutes for model revision.] Circulate and ask probing questions: 'Kamau, your backtracker's velocity goes	Cumulative model revision (Iterative Modelling). By returning to the same anchoring phenomenon model each lesson and adding a new mathematical layer, learners build a coherent, interconnected understanding rather than isolated skills. Requiring a written caption forces consolidation of graphical, algebraic, and conceptual understanding into a single unified narrative.	Observable indicators: (1) Model correctly shows a roughly horizontal v–t line for the straight-line walker, an irregular line for the zigzag walker, and a line that crosses zero (into negative velocity) for the backtracker. (2) Acceleration arrows correctly point in the direction of velocity change (or are labelled zero for constant sections). (3) Three-sentence caption uses all four target words correctly and connects to the Timer Race Puzzle. Exit check: collect one model caption per table group as evidence of lesson-level

[Over Time (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	three sentences explaining how their model now shows both velocity and acceleration — and why two objects with the same speed but different accelerations will end up in different places.	negative — what does that mean for their acceleration when they turn around? How does that explain why they end up in a different place?' [WAIT 12 seconds.] 'Aisha, can you show me on your model where acceleration is zero?' [WAIT 10 seconds.] With 3 minutes remaining: 'Write your three-sentence model caption. It must include the words: displacement, velocity, acceleration, and gradient.' Invite 2 volunteers to read their captions aloud. Close with: 'Next lesson, we will discover what the AREA under a v–t graph tells us — which will give us one more superpower for predicting motion.'		understanding.

D. TEACHER REFLECTION
After the lesson, consider: (1) Could learners correctly distinguish between the HEIGHT of a v–t graph (velocity) and the SLOPE (acceleration) — or did confusion persist? Note which pairs needed repeated re-explanation and plan a targeted warm-up for Lesson 8. (2) Were the Kipchoge data values realistic enough to feel genuine while being clean enough to calculate easily? If learners struggled with the numbers, consider pre-rounding values on Handout 7A. (3) Did the DQB update reveal any persistent misconceptions about direction (vector nature of velocity and acceleration) that need to be addressed before Lesson 8 on area under v–t graphs? (4) Which learners are still struggling to choose appropriate scales for their graphs? Identify these learners for additional graph-skills scaffolding. (5) Did the connection back to the Timer Race Puzzle feel genuine and motivated, or did it feel forced? Adjust the matatu / Kipchoge context ratio based on which resonated more strongly with your class.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners plotted velocity–time graphs from tabulated data for Eliud Kipchoge's marathon training splits and a Nairobi–Thika matatu journey. They observed that different sections of the v–t graph (horizontal, rising, falling) correspond to different types of motion.
What did I learn?	The gradient of a velocity–time graph equals the acceleration of the object ($a = \Delta v \div \Delta t$). A horizontal line means zero acceleration (uniform velocity); a positive slope means positive acceleration (speeding up); a negative slope means deceleration (slowing down). Acceleration is a vector quantity — a negative value indicates a change in direction or slowing down.
How does this explain the phenomenon?	Learners used v–t graphs and gradient calculations to explain how the three Timer Race characters (straight-line walker, zigzag walker, backtracker) have different accelerations even when covering similar distances, and why knowing velocity alone is not enough — the rate of change of velocity (acceleration) and direction together determine where an object ends up.

LESSON 8 (80 minutes): Interpreting Velocity–Time Graphs & Area Under the Graph	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To interpret velocity–time graphs, calculate acceleration from the gradient, and discover that the area under a velocity–time graph equals displacement — applying these skills to the boda-boda braking scenario.
Knowledge	Learners will know that: (1) the gradient of a velocity–time graph gives acceleration; (2) the area under a velocity–time graph gives displacement; (3) positive and negative gradients represent acceleration and deceleration respectively; (4) a horizontal line on a v–t graph represents uniform velocity (zero acceleration).
Skills	Learners will be able to: (1) read and interpret velocity–time graphs including segments of acceleration, uniform velocity, and deceleration; (2) calculate acceleration using the gradient of a v–t graph; (3) calculate displacement by finding the area under a v–t graph (rectangles and triangles); (4) apply these calculations to determine stopping distances in a real-world boda-boda braking scenario.
Attitudes	Learners will appreciate that mathematical tools such as graphs are powerful and precise ways of describing real-world motion, and will value careful, direction-aware thinking when interpreting motion data.
Key Inquiry Question	How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Purpose in Storyline	In Lesson 7, learners plotted displacement–time graphs and extracted velocity from gradients. Lesson 8 advances the sequence by moving to velocity–time graphs, where learners discover that gradients now encode acceleration and enclosed areas encode displacement. This unlocks the ability to answer the driving question fully: knowing velocity alone is not enough — we must track how velocity changes with time and in which direction to predict exactly where a moving object ends up. The boda-boda braking scenario makes the stakes concrete: how much road does a boda-boda need to stop safely?
Safety Notes	No physical hazards anticipated. If learners re-enact slow walking activities in the classroom or corridor to generate data, ensure the walkway is clear of bags and obstacles. Remind learners to walk — not run — during any data-collection movement.

B. LESSON OVERVIEW
Learners open with a brief recall of Lesson 7 displacement–time graphs, then shift focus to velocity–time graphs. In the Predict phase, they sketch what they think a v–t graph looks like for a boda-boda that accelerates away from a stage, rides at constant speed, then brakes sharply to a stop. In the Observe phase, learners work in groups to analyse four printed v–t graph cards representing different motion scenarios (including the boda-boda) and answer guided questions. In the Explain phase, the teacher leads a whole-class discussion formalising the gradient = acceleration relationship and facilitating a guided discovery that the area of the shapes under the graph equals displacement. Learners then calculate stopping distance for the boda-boda. In the DQB phase, new evidence cards are added to the class Driving Question Board. Finally, in the Model Building phase, learners update their cumulative motion models to incorporate v–t graph representations alongside their earlier d–t graph representations.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are presented with a	Distribute the scenario card face-	Elicit–Confront–Resolve	Observable indicator: Learner

📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	scenario card: 'A boda-boda rider at a stage in Kisumu accelerates from rest to 30 km/h over 10 seconds, travels at constant 30 km/h for 20 seconds, then brakes to a complete stop in 5 seconds.' Without any instruction, learners individually sketch what they think this motion looks like on a velocity–time graph (v on y-axis, t on x-axis) in their exercise books. They label each segment with a word (e.g., 'speeding up', 'steady', 'slowing down') and write one sentence predicting: 'I think the area of each section of the graph might represent…'	down. Say: 'Before we look at any graphs today, I want you to draw what you THINK the graph looks like — this is your prediction, so there are no wrong answers yet. Flip your card over. You have 4 minutes. Go.' [Monitor silently — do NOT correct or hint. After 4 minutes:] 'Now turn to your partner. Compare your sketches. Where are they the same? Where are they different? 2 minutes.' [After pair discussion — cold-call 3 pairs, deliberately choosing one sketch that shows a curved line, one that shows straight lines, and one that shows a horizontal segment.] Ask each: 'Walk us through your thinking — why did you draw it that way?' [WAIT 12 seconds after each student begins speaking before prompting.] Record 2–3 contrasting prediction sketches on the board WITHOUT evaluating them. Say: 'We will come back to these at the end of class and see which prediction the evidence supports.'	(Prediction Sketching): By committing a sketch to paper before instruction, learners activate prior knowledge from Lesson 7 (d–t graphs) and expose their current mental models. Contrasting predictions on the board create productive cognitive dissonance that motivates careful evidence gathering.	sketches show at least three distinct segments. Listen for whether learners distinguish between 'how fast' (slope of d–t) and 'the speed itself' (y-value on v–t) — this reveals whether the Lesson 7 gradient concept has transferred. Note any learner who draws a curved line for constant acceleration — this is a key misconception to address in the Explain phase.
Observe Phase 📹 **VIDEO:** [AP Physics 1 review of Forces and Newton's Laws](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Line Graphs to Display Data Over Time (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar	In groups of 4, learners receive a printed 'Graph Card Set' containing four velocity–time graphs (Graph A: uniform velocity; Graph B: uniform acceleration from rest; Graph C: acceleration then deceleration back to rest — the boda-boda journey; Graph D: a matatu that accelerates, travels at constant speed, decelerates, stops briefly, then reverses — shown as negative velocity). Each group also receives a structured 'Graph Investigation Sheet' with the following guided questions for each graph: (1) Describe the motion in words. (2) Calculate the gradient of each segment — show your working. (3) What does this	Before distributing cards, say: 'Your job right now is NOT to get the right answer. Your job is to gather evidence from these graphs. Think like a detective — what is the graph showing you?' [Distribute graph card sets.] Circulate continuously. At each group, crouch to their level and use targeted prompts rather than answers: 'What are the units on each axis? What do you get when you multiply those units together?', 'What does a steeper line tell you compared to a gentler line?', 'Look at Graph D — the velocity goes negative. In the context of the matatu, what does negative velocity mean?' [WAIT 10 seconds	Structured Group Investigation with Unit Analysis: By calculating gradient units (km/h ÷ s = km/h/s, i.e. acceleration) and area units (km/h × s = km — with unit conversion) from the graph axes, learners derive the physical meaning of gradient and area through dimensional reasoning rather than being told. This embeds the NGSS practice of Using Mathematics and Computational Thinking.	Observable indicators: (1) Learners correctly multiply axis units to identify area units (at least 3 of 4 groups). (2) Gradient calculations show rise-over-run working with correct sign for deceleration. (3) Listen for learners connecting Graph D's negative velocity to the 'direction matters' thread from the Timer Race Puzzle phenomenon — this indicates transfer of the unit's core idea. Flag any group that calculates area of a triangle as base × height (forgetting the ½) for targeted correction in Explain phase.

readings	gradient tell you about the motion? (4) Shade the area under each segment. Estimate the total shaded area. What unit would this area have (km/h × seconds)? (5) What do you think this area might represent physically? Groups discuss and record answers collaboratively. All four learners must write in their own books.	after each prompt before adding more.] After 15 minutes of group work, pause the class: 'Freeze. I want one person from each group — not the same person who always talks — to tell me: what unit did you get for the area, and what do you think it represents? Group 1 — go.' [Cold-call all 4 groups in sequence. Record their area-unit answers on the board in a table: Group | Unit calculated | Physical meaning proposed.] Do not confirm or deny yet. Say: 'Hold that thought. We are about to find out together.'		
Explain Phase 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes for a teacher-facilitated whole-class discussion. Learners first share and defend their group findings. The teacher then formalises two key relationships through worked examples: (1) Gradient of v–t graph = (Δv ÷ Δt) = acceleration (with units km/h/s or m/s²). (2) Area under v–t graph = displacement. Learners copy a structured summary into their books and immediately apply both relationships to the boda-boda braking problem: given the boda-boda decelerates from 30 km/h to 0 in 5 seconds, learners calculate (a) the deceleration and (b) the stopping displacement (area of triangle under the graph). A second challenge problem introduces a Nairobi matatu: 'A matatu accelerates from rest at 2 m/s² for 8 seconds, then brakes uniformly to rest in 4 seconds. Calculate total displacement.' Learners solve individually, then compare answers with a partner.	Begin by returning to the board predictions from Phase 1: 'Let's look at our predictions. [Point to each sketch.] Which one matches the evidence from your graph cards?' [Cold-call 2 learners — WAIT 12 seconds.] Then say: 'I am now going to show you why the gradient and the area are so powerful.' [Write on board: GRADIENT of v–t graph = Δv/Δt = acceleration. Walk through the boda-boda Graph C gradient calculation step by step, narrating every line: 'Gradient = (30 – 0) km/h ÷ 10 s = 3 km/h per second. This means every second the boda-boda's velocity increases by 3 km/h. That is acceleration.'] Then say: 'Now the area. What shape is under the acceleration segment? A triangle. Area = ½ × base × height = ½ × 10 s × 30 km/h = 150 km·s/h.' [Pause.] 'Does that unit make sense for displacement? Let's convert: 150 km·s/h × (1 h / 3600 s) = 0.0417 km = 41.7 m. So the boda-boda travelled about 42 metres while accelerating.' [Write this on the board clearly.] Ask: 'What does this mean for the	Concept Formalisation with Worked Example Anchoring: The teacher bridges from learners' own unit-analysis discoveries (Phase 2) to formal definitions, ensuring the abstraction is grounded in what learners already found. The boda-boda context keeps the mathematics tethered to the phenomenon throughout, exemplifying the NGSS practice of Constructing Explanations.	Observable indicators: (1) Learners calculate deceleration of the braking boda-boda as −6 km/h/s (or −1.67 m/s²) with correct negative sign. (2) Stopping displacement calculated as ½ × 5 × 30 = 75 km·s/h = 20.8 m ≈ 21 m. (3) Matatu challenge: learners find displacement during acceleration = ½ × 8 × 16 = 64 m and during braking = ½ × 4 × 16 = 32 m, total = 96 m. Spot-check the 6 collected books for correct triangle area formula, correct unit conversion, and correct sign convention for deceleration.

		braking triangle — the 5-second stop? Calculate it with your partner now. You have 3 minutes.' [WAIT — circulate — cold-call one pair to present working on the board.] For the matatu challenge problem, say: 'Work this one alone first — 5 minutes. This is your chance to show ME what you understand.' [Collect 6 books for rapid formative check while learners compare with partners.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Individually, each learner writes two sticky-note style index cards: (1) an EVIDENCE card — one thing they can now prove from a v–t graph (gradient = acceleration OR area = displacement), written as a claim with evidence; (2) a NEW QUESTION card — one new question or wonder this lesson has raised for them. Learners post both cards on the class Driving Question Board in the designated Lesson 8 column. The class does a 90-second silent gallery walk of the new cards before sitting down.	Say: 'We have been building evidence all unit to answer our driving question: How can mathematics help us fully describe, graph, and predict the motion of objects? Today you found two powerful new tools. Write your evidence card in this format: I CAN CLAIM that [statement] BECAUSE [evidence from today's graph work].' [Model one example on the board: 'I can claim that the boda-boda needed at least 21 metres to stop BECAUSE the area of the triangle under the v–t graph equals displacement.'] 'You have 3 minutes to write both cards.' [After posting:] 'Walk around and read at least 4 other cards — look for any evidence claims that surprise you or questions that match yours.' [After gallery walk, cold-call 2 learners: 'Whose evidence card caught your eye and why?' WAIT 10 seconds.]	Claim–Evidence–Reasoning (CER) Card Writing: Framing the DQB update as a CER task trains learners to distinguish between statements of fact and evidence-backed claims — a core scientific practice. The gallery walk creates peer accountability and surfaces shared questions that can guide Lesson 9.	Observable indicator: Evidence cards contain a specific quantitative claim (e.g., a calculated value) linked to graph evidence — not just a definition. Question cards reveal whether learners are thinking ahead (e.g., 'What if the velocity changes in a curved way, not a straight line?' — foreshadowing non-uniform acceleration). Collect cards after class to identify the 2–3 most common new questions for incorporation into Lesson 9's opening.
Model Building Phase 📹 **VIDEO:** AP Physics 1 review of Forces and Newton's Laws	Learners retrieve their cumulative 'Motion Model' — a visual representation they have been building since Lesson 1, showing how different concepts connect. Today they add a new layer: a small v–t graph sketch of the boda-boda journey, annotated with	Say: 'Open your cumulative motion models. You have been building this since Lesson 1. Today we are adding the v–t graph layer.' [Display a partially completed model on the board as a scaffold — show the blank space where learners should add today's	Cumulative Model Revision (Visual Knowledge Integration): The running model forces learners to explicitly connect new concepts (v–t graphs, acceleration, area = displacement) to previously established ideas (d–t graphs, velocity as gradient). Drawing the	Observable indicators: (1) The connector arrow between d–t and v–t graphs is present and correctly labelled. (2) The boda-boda v–t sketch shows three distinct segments with correct shapes (rising line, horizontal, falling line to zero). (3) The phenomenon

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Line Graphs to Display Data Over Time (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	arrows pointing to the gradient (labelled 'acceleration') and the shaded area (labelled 'displacement'). They also draw a connector arrow from their earlier d–t graph model (Lesson 7) to today's v–t graph, labelling the arrow: 'The slope of d–t gives velocity; the slope of v–t gives acceleration.' At the bottom of their model, learners write one sentence connecting today's learning back to the Timer Race Puzzle: 'This connects to our puzzle because…'	content, but do NOT pre-fill it.] 'Add your boda-boda v–t sketch, label the gradient and the area, draw your connector arrow to the d–t graph, and write your phenomenon connection sentence. You have 8 minutes.' [Circulate — target learners who struggled in Phase 3 and check that their area labels are correct.] With 2 minutes remaining, cold-call one learner to read their phenomenon connection sentence aloud. Ask the class: 'Does that connection make sense to you? Thumbs up, sideways, or down.' [Address any sideways or down thumbs with a brief clarifying question before closing.]	connector arrow is a metacognitive act — learners must articulate the relationship between the two graph types, consolidating the conceptual hierarchy of displacement → velocity → acceleration. This reflects the NGSS practice of Developing and Using Models.	connection sentence references direction, location, or the Timer Race Puzzle specifically — not just motion in general. Photograph 4–5 models for display and for informing Lesson 9 planning.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners arrive at the area = displacement relationship through their own unit analysis (Phase 2), or did groups require heavy prompting — if so, how might the Graph Investigation Sheet be redesigned to scaffold the unit-multiplication step more clearly? (2) Which of the four graph cards proved most challenging? Graph D (negative velocity) often exposes unresolved confusion about direction from earlier lessons — note which learners struggled and plan a brief re-engagement in Lesson 9's opening. (3) Were learners able to apply the area formula correctly to triangles AND rectangles, or did errors cluster around one shape type? (4) Review the DQB question cards collected after class — do the new questions suggest learners are ready for non-uniform acceleration and curved v–t graphs in Lesson 9, or is there a need to consolidate today's linear (uniform) cases first? (5) Did the boda-boda context feel authentic and motivating, or did learners need a stronger personal connection — consider inviting learners to estimate stopping distances for vehicles near their own homes for a homework extension.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners analysed four velocity–time graph cards representing different motion scenarios (including a boda-boda journey and a matatu trip in Nairobi). They calculated gradients of each segment, identified areas of rectangles and triangles under the graphs, and multiplied axis units to determine what each area physically represents.
What did I learn?	The gradient of a velocity–time graph equals acceleration ($\Delta v \div \Delta t$), with a negative gradient indicating deceleration. The area under a velocity–time graph equals displacement. Using these two relationships, learners calculated that the boda-boda required approximately 21 metres to stop from 30 km/h, and that a Nairobi matatu covered 96 metres total in a combined acceleration-and-braking sequence.
How does this explain the phenomenon?	Learners explained why direction matters on a v–t graph by interpreting negative velocity on Graph D as the matatu reversing — connecting directly to the Timer Race Puzzle, where objects covering the same numerical distance can end up in completely different locations depending on direction. The area under the v–t graph gives the actual displacement — not just how far, but in

	which direction — which is exactly the richer mathematical language the driving question demands.

LESSON 9 (80 minutes): Relative Speed — Combining Velocities When Direction Matters	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to understand and apply the concept of relative speed by combining velocities of objects moving in the same or opposite directions along a straight line.
Knowledge	Learners will know that relative speed is the speed of one object as observed from another moving object; that objects moving in the same direction have a relative speed equal to the difference of their speeds; and that objects moving in opposite directions have a relative speed equal to the sum of their speeds.
Skills	Learners will be able to derive and apply the rules for relative speed in same-direction and opposite-direction scenarios; solve numerical problems involving two objects on a straight-line path; and represent relative motion situations using annotated diagrams.
Attitudes	Learners will appreciate the importance of specifying direction when describing motion, demonstrate patience and precision when combining vector quantities, and show curiosity in connecting mathematical rules to real-world Kenyan transport contexts.
Key Inquiry Question	How do distance, displacement, speed, velocity, and acceleration differ from one another, and how are they calculated? How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Purpose in Storyline	In Lesson 9, learners have already built understanding of displacement, velocity, acceleration, and distance-time and velocity-time graphs. This lesson extends vector thinking by asking: what happens when TWO moving objects are considered together? By studying two matatus on the Nairobi–Mombasa highway, learners discover that direction determines whether velocities add or subtract — deepening the unit's central argument that direction always matters. This directly advances the driving question and prepares learners for the final synthesis lesson.
Safety Notes	This is a reading and illustration-based lesson conducted entirely in the classroom. No physical movement or equipment is required. Ensure learners handle printed materials carefully. If the lesson moves outdoors for any brief demonstration, learners must remain on the footpath and not near the road.

B. LESSON OVERVIEW
Learners begin by revisiting the anchoring phenomenon — recalling how the same numerical speed meant different things for zigzag and straight-line walkers — and predict what a passenger inside a moving matatu would observe about another matatu on the road. They then read a structured passage on relative speed accompanied by two illustrated scenarios: (1) two matatus travelling in the same direction on the Nairobi–Mombasa highway, and (2) two matatus travelling toward each other. Through guided annotation of the illustrations and worked examples, learners derive the two core rules: relative speed (same direction) = $v_1 - v_2$, and relative speed (opposite direction) = $v_1 + v_2$. They solve a set of Kenya-contextualised problems, update the Driving Question Board with newly answered questions, and revise their cumulative motion models to incorporate the idea of relative velocity.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are presented with the	Write the scenario on the board	Prediction Polling with Justification	Observable indicator: Can the

📹 **VIDEO:** [Mean value theorem application](#) Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos](#) 📖 **HTML:** [Graphing when the Degrees of the Numerator and Denominator are the Same (Flexbook)](#) Source: CK-12 [🔍 Search ARES for similar readings](#)	following written scenario on the board: 'You are a passenger in Matatu A travelling from Nairobi to Mombasa at 80 km/h. Matatu B is also travelling from Nairobi to Mombasa at 100 km/h and is directly behind you. From YOUR window, how fast does Matatu B appear to be approaching you?' Learners write an individual prediction in their exercise books, choosing from: (a) 180 km/h, (b) 100 km/h, (c) 20 km/h, (d) 0 km/h — and write one sentence justifying their choice. They then share with a shoulder partner for 2 minutes before the teacher takes a class poll by show of hands.	before learners enter the room. Once settled, say: 'Before we read anything today, I want you to trust your instincts. Read this scenario silently and make your prediction — no calculators, no discussion yet. You have 90 seconds.' [Allow 90 seconds of silent individual thinking.] After writing, say: 'Now turn to your shoulder partner. Share your prediction AND your reason. You have 2 minutes.' [Circulate, listening for the range of reasoning — do not correct yet.] After 2 minutes, conduct a show-of-hands poll: 'Hands up for (a) 180 km/h… (b) 100 km/h… (c) 20 km/h… (d) 0 km/h.' Record the tally on the board visibly. Cold-call THREE learners — one from the majority answer, one from a minority answer, one who hesitates — asking each: 'Tell us your reasoning in one sentence.' [Allow 10 seconds WAIT TIME after each cold call before accepting the answer.] Do NOT confirm or deny any prediction. Say: 'Hold these ideas. Our reading today will give you the evidence to judge your own prediction.'	— By forcing a choice among four specific numerical answers and requiring a written justification, learners activate prior knowledge about velocity and direction. The visible tally of disagreement creates productive cognitive tension that motivates reading. Connecting back to the phenomenon: just as the zigzag and straight-line walkers had the same numerical speed yet different experiences, two matatus at different speeds create a new puzzle about what 'speed' means to an observer inside one of them.	learner articulate whether direction is relevant to their prediction? Listen for learners who say 'it depends on which way they are going' — this signals readiness for the lesson. Flag learners who choose (b) 100 km/h with no reference to Matatu A's speed, as they may conflate absolute and relative speed. Record the class poll tally to revisit at the end of the lesson.
Observe Phase 📹 **VIDEO:** [Mean value theorem application](#) Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos](#) 📖 **HTML:** [Graphing when the Degrees of the Numerator](#)	Learners receive a two-page printed reading passage titled 'Two Matatus on the Nairobi–Mombasa Highway' (or view it on-screen via the ARES platform). Page 1 contains the passage with definitions and two fully labelled illustrations: Illustration A shows Matatu A (80 km/h) and Matatu B (100 km/h) both moving in the same direction (east, toward Mombasa), with arrows of proportional length; Illustration B shows Matatu C (90 km/h) moving east and Matatu D (70 km/h)	Distribute the reading or direct learners to it on-screen. Say: 'This is a reading-and-illustration lesson. Your job is to be a detective — use three annotation symbols as you read: tick for understood, question mark for confused, star for surprised. Read silently for 8 minutes. Do not skip the illustrations — spend at least 2 minutes on each one.' [Set a visible timer for 8 minutes. Circulate quietly, noting which illustrations learners linger on.] After 8 minutes, say: 'Now open	Annotation Reading with Illustration Analysis — Structured annotation (✓ ? ★) ensures active reading rather than passive scanning. The Annotation Table forces learners to compare the two scenarios side-by-side, making the contrast between same-direction (subtraction) and opposite-direction (addition) concrete and memorable. Tracing worked examples with a pencil activates procedural engagement. Rephrasing rules in their own words promotes deeper encoding.	Observable indicator: Does the learner's Annotation Table correctly contrast the two illustrations — specifically identifying that in Illustration A the relative speed is 100 − 80 = 20 km/h and in Illustration B it is 90 + 70 = 160 km/h? Listen for learners who write 'relative speed is always the bigger number minus the smaller number' — this is a misconception; they have missed the role of direction. Spot-check 5 Annotation Tables during circulation. Ask targeted learners:

and Denominator are the Same (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	moving west (toward Nairobi), with arrows pointing in opposite directions. Page 2 contains two worked examples with full solutions. Learners first read silently for 8 minutes, annotating with: a tick (✓) for things they understood, a question mark (?) for things that confused them, and a star (★) for things that surprised them. They then complete a structured Annotation Table in their books with three columns: 'What I noticed in Illustration A', 'What I noticed in Illustration B', 'How this connects to the phenomenon.'	your exercise book to a fresh page. Draw three columns: Illustration A observations, Illustration B observations, and Connection to the phenomenon. Fill these in — you have 5 minutes.' [Circulate and cold-call two learners to share one observation each from their table. Allow 10 seconds WAIT TIME per learner.] Then say: 'Look at the worked examples on Page 2. Trace through each step with your pencil — make sure you can see where every number comes from.' [Allow 4 minutes for this.] Finally, say: 'In your own words, write the two rules you just read as a single sentence each. Do not copy — rephrase.'	Connection to phenomenon: learners are asked explicitly to link each illustration back to the timer race puzzle — just as the zigzag walker and straight-line walker experienced the same distance differently, two matatus at the same speed in different directions create entirely different relative motion experiences.	'Why do we subtract in one case and add in the other?' [Wait 10 seconds.]
Explain Phase 📹 **VIDEO:** Mean value theorem application Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Graphing when the Degrees of the Numerator and Denominator are the Same (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Working in groups of four, learners complete a problem set of five relative-speed questions, all set in Kenyan contexts. Problems include: (1) A Nairobi SGR train travelling at 120 km/h overtakes a matatu at 80 km/h moving in the same direction — find the relative speed of the train as seen from the matatu. (2) Two buses leave Kisumu and Eldoret simultaneously toward each other at 90 km/h and 70 km/h — how fast are they approaching each other? (3) Eliud Kipchoge runs at 20 km/h; a cyclist travelling in the same direction passes him at 30 km/h — what is the relative speed of the cyclist as observed by Eliud? (4) A boda boda at 40 km/h and a lorry at 60 km/h travel toward each other on the Thika Superhighway — find their relative speed of approach. (5) Challenge: A matatu travelling at v_1 km/h is overtaken by a car at v_2 km/h in the same direction. Write a general	Assign groups before distributing the problem set. Say: 'In your groups of four, solve all five problems. Show ALL working — units, direction arrows on diagrams, and the rule you used. You have 15 minutes.' [Set a visible timer. Circulate actively — do not answer content questions directly; instead ask redirecting questions: 'What direction is each object moving?' 'Which rule from your reading applies here?' 'Draw arrows first before writing any numbers.'] After 15 minutes, cold-call two groups (not the fastest finishers) to present Question 5 verbally. Say to the presenting group: 'Tell us: what is the general rule, and what does it mean physically when v_1 equals v_2?' [Allow 10 seconds WAIT TIME before the group responds.] After presentations, pose to the whole class: 'Return to your prediction from Phase 1. Were you right? Which rule applies to the matatu	Collaborative Problem Solving with Peer Presentation — Group problem solving allows learners to negotiate meaning and correct each other's direction errors in real time. The algebraic generalisation in Question 5 elevates the activity from procedural to conceptual, requiring learners to see the rule as a mathematical structure rather than a recipe. The return to the Phase 1 prediction closes the predict-observe-explain loop, making the learning arc visible to learners. Phenomenon connection: the teacher explicitly names the link between the matatu relative-speed result and the original timer race puzzle, reinforcing the unit's central argument.	Observable indicator: Does the learner correctly identify which rule (addition or subtraction) applies based on direction, not on which speed is larger? A learner demonstrating mastery will draw directional arrows on their diagram before writing any formula. Watch for groups that always subtract the smaller from the larger regardless of direction — this indicates direction is not yet understood as the determining factor. During presentations, assess whether the group can articulate verbally: 'When objects move in opposite directions, the relative speed is the sum because they are closing the gap from both sides.'

	algebraic expression for the relative speed of the car as seen from the matatu, and state what happens if $v_1 = v_2$. After solving, each group prepares a 2-minute verbal explanation of Question 5, using the rule they derived from the reading.	scenario — addition or subtraction, and why?' [Allow 15 seconds WAIT TIME, then cold-call THREE learners who were in the minority prediction group.] Confirm the correct answer: 100 − 80 = 20 km/h, because both matatus move in the same direction. Connect explicitly: 'This is exactly why direction always matters — the same two speeds, in different directions, give completely different results, just like our timer race puzzle.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Mean value theorem application] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Graphing when the Degrees of the Numerator and Denominator are the Same (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. Each learner receives two sticky notes: one green, one yellow. On the GREEN note, they write one question from the DQB that today's lesson has fully answered (e.g., 'How does direction affect the speed we observe between two moving objects?'). On the YELLOW note, they write one new question that today's lesson has raised or that they still cannot answer. Learners post both notes on the DQB in the appropriate columns ('Answered' and 'Still Wondering'). The class briefly reviews which questions are migrating from 'wondering' to 'answered' as the unit nears its end.	Distribute sticky notes. Say: 'We are now at Lesson 9 of 10. Look at the DQB together. Take 3 minutes silently — find one question this lesson answered, and think of one new question it raised or one you still cannot answer.' [Allow 3 minutes of silent reflection.] After learners post their notes, read aloud two or three of the newly answered questions and ask: 'Who agrees this question is now answered? Thumbs up if you can explain the answer in one sentence.' Cold-call ONE learner per question. [Wait 10 seconds per cold call.] Then draw attention to the yellow 'still wondering' notes. Say: 'These yellow notes are not failures — they are the engine of tomorrow's final lesson. Look at what we still don't know.' Point to any DQB questions about graphing both objects simultaneously or predicting when two objects meet — say: 'Lesson 10 will help us close these.' Record a count of answered vs. unanswered questions on the board for learner visibility.	DQB Migration with Metacognitive Reflection — Moving questions from 'wondering' to 'answered' makes the growth of understanding visible and concrete, reinforcing a sense of progress. The requirement to write a new yellow question ensures learners remain in inquiry mode rather than assuming they are 'done.' Tracking the ratio of answered to unanswered questions across the unit builds metacognitive awareness of their own learning trajectory. Phenomenon connection: encourage learners to note whether any DQB questions connect back to the original timer race — e.g., 'If the zigzag walker and straight-line walker had been moving toward each other, what would their relative speed be?'	Observable indicator: Is the learner's green note specific and accurate — does it name a concept (relative speed, direction-dependent velocity combination) rather than a vague statement like 'I learned about speed'? Listen for yellow notes that anticipate Lesson 10 content (e.g., 'How do we graph two objects on the same distance-time graph?' or 'How do we find when two objects meet?') — these indicate strong forward-thinking engagement. Flag learners who cannot identify any answered question, as they may need targeted support before Lesson 10.
Model Building	Learners open their cumulative motion model — the annotated	Say: 'Open your cumulative model. Today we are adding a new	Cumulative Model Revision with Annotated Diagrams — The	Observable indicator: Does the learner's updated model correctly

Phase 📹 **VIDEO:** Mean value theorem application Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Graphing when the Degrees of the Numerator and Denominator are the Same (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	diagram or concept map they have been building since Lesson 1. They add a new section labelled 'Relative Velocity' and update it with: (1) a two-arrow diagram showing same-direction motion with the label 'Relative speed = $v_1 - v_2$'; (2) a two-arrow diagram showing opposite-direction motion with the label 'Relative speed = $v_1 + v_2$'; (3) a written note connecting this to the unit's central claim: 'Direction determines whether velocities combine by addition or subtraction.' Learners also draw a small sketch of the Nairobi–Mombasa highway scenario from their reading and annotate it with the relevant values from the worked example. Finally, they write a one-sentence 'model update statement': 'My model now shows that…'	branch — relative velocity. This is not a separate idea; it is a direct extension of everything you have already built about vectors and direction. Take 8 minutes to update your model using the three items I'll display.' [Display the three update items on the board.] Circulate and look specifically for whether learners are drawing arrows in opposite directions for the opposite-direction scenario — not just writing numbers. After 8 minutes, cold-call TWO learners to read their 'model update statement' aloud. [Allow 10 seconds WAIT TIME per learner.] After both share, say: 'Notice that both of these statements include the word direction. That is not a coincidence — that is the answer to our driving question. As we head into our final lesson tomorrow, your model should now be rich enough to describe, graph, and predict motion. Ask yourself: what is the one piece your model still needs?' [Allow 30 seconds of silent reflection before dismissing.]	physical act of adding to a model built over nine lessons makes knowledge integration visible. Requiring both a symbolic rule (algebraic expression) and a visual representation (arrow diagram) ensures learners encode the concept in multiple modalities. The 'model update statement' is a metacognitive prompt that forces the learner to articulate change — not just add content, but notice what is new. Phenomenon connection: the final reflection question — 'What is the one piece your model still needs?' — is deliberately designed to make learners feel the incompleteness of their model, priming them for Lesson 10's final synthesis.	show arrows in opposite directions for the opposite-direction scenario, rather than two arrows pointing the same way? Does the algebraic label use $v_1 - v_2$ for same direction and $v_1 + v_2$ for opposite directions? A model that correctly uses arrows and algebra together signals mastery. A model that only writes words without diagrams or only writes numbers without direction indicators signals a gap. Collect or photograph 6–8 models at random to review before Lesson 10 for targeted follow-up.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) When you conducted the Phase 1 prediction poll, what was the distribution of answers — and did the majority answer shift by the end of Phase 3? If not, what additional bridge was needed between the reading and the correction of the prediction? (2) During Phase 2 annotation, did learners spend adequate time on the illustrations, or did they skip to the text? If they skipped, consider adding a 'illustration-first' protocol in future — cover the text and ask learners to describe what they see in the arrows before reading. (3) In Phase 3 group work, did any groups consistently subtract the smaller speed from the larger regardless of direction? If so, design a targeted 'direction-first' prompt card for Lesson 10 revision. (4) Review the DQB yellow notes: are there clusters of unanswered questions about graphing two objects simultaneously or finding meeting points? These should anchor the Lesson 10 opening. (5) Photograph 6–8 learner models from Phase 5 and check whether the arrow diagrams correctly distinguish same-direction from opposite-direction scenarios — this is the key conceptual indicator for readiness for the final synthesis lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners read an illustrated passage showing two matatus on the Nairobi–Mombasa highway — one pair travelling in the same

	direction and one pair travelling in opposite directions — and annotated the diagrams to compare relative speeds in each scenario.
What did I learn?	Learners derived and applied two relative-speed rules: when two objects move in the same direction, relative speed = $v_1 - v_2$; when they move in opposite directions, relative speed = $v_1 + v_2$. They solved Kenya-contextualised problems and generalised the rule algebraically.
How does this explain the phenomenon?	Direction determines whether two velocities combine by addition or subtraction. This extends the unit's central principle — that direction always matters — to situations involving two moving objects, deepening the answer to the driving question about why direction must always be specified when describing motion.

LESSON 10 (80 minutes): Application & Cumulative Model Building: The Final Explanation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all concepts of linear motion — distance, displacement, speed, velocity, and acceleration — by applying them to a rich real-world problem and constructing a final, comprehensive mathematical model of motion.
Knowledge	Learners recall and integrate: (1) velocity as rate of change of displacement ($v = s/t$), (2) acceleration as rate of change of velocity ($a = (v-u)/t$), (3) scalar vs. vector distinction, (4) motion graph interpretation (distance-time and velocity-time), and (5) equations of uniform acceleration. They articulate how direction transforms a scalar into a vector and why this distinction matters in real-world prediction.
Skills	Learners apply multi-step calculations involving velocity, acceleration, and displacement to a real-life highway overtaking scenario; construct and annotate a complete velocity-time graph; compare their Lesson 1 naive model with their final model; and present mathematical reasoning clearly in written and oral form.
Attitudes	Learners demonstrate perseverance in tackling complex, multi-step problems; appreciate the power of mathematics as a tool for safety and decision-making in everyday Kenyan life; and show intellectual honesty by openly comparing what they believed at the start of the unit with what the evidence has taught them.
Key Inquiry Question	How do distance, displacement, speed, velocity, and acceleration differ from one another, and how are they calculated? How can graphs be used to represent and interpret the motion of objects moving in a straight line?
Purpose in Storyline	This is the culminating lesson of the unit. Learners bring every concept and skill developed across Lessons 1–9 to bear on a single, rich real-world problem — the safe overtaking of a truck on the Nakuru–Nairobi highway. By solving this problem, finalising their cumulative motion models, and comparing them with their Lesson 1 models, learners demonstrate full progression from the naive puzzle of the timer race to confident, precise mathematical description and prediction of linear motion. The driving question is answered completely and the DQB is closed.
Safety Notes	No physical activity or materials hazards in this lesson. Ensure learners are reminded that the highway scenario discussed is a mathematical model for learning purposes only — safe driving behaviour and road safety are serious real-world responsibilities. Remind learners not to attempt to perform or encourage dangerous overtaking manoeuvres in real life.

B. LESSON OVERVIEW

Lesson 10 is the final lesson of Sub-Strand 3.4: Linear Motion. It opens by returning to the anchoring phenomenon (the Timer Race Puzzle) and to the Driving Question Board (DQB) accumulated across all nine previous lessons, asking learners to assess how far their understanding has grown. The core activity is a rich, multi-part real-world problem: planning the mathematically safe overtaking of a truck on the Nakuru–Nairobi highway. Learners must integrate velocity, acceleration, displacement, and graphical representation to solve the problem and justify whether the overtake is safe. Following this, learners finalise their cumulative annotated motion models — begun in Lesson 1 and revised across every lesson — and present a side-by-side comparison of their Lesson 1 naive model and their final model, narrating the evidence journey. The lesson closes with a whole-class DQB completion ceremony, a Final Explanation discussion tying every concept back to the anchoring phenomenon, and individual written exit reflections.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Mean value theorem application] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the original Timer Race Puzzle scenario from Lesson 1 — projected on the board or read aloud — and are given 3 minutes to silently re-read their very first written prediction from Lesson 1 (kept in their science journals). They then individually write a brief 'then vs. now' note: What did I think then? What do I know now? Next, learners view the full DQB — all questions posted across Lessons 1–9 — and in pairs spend 4 minutes categorising each sticky note or DQB entry as: (A) Fully answered, (B) Partially answered, or (C) Still uncertain. Pairs report their categorisations to the class. The teacher records consensus on the board. Learners then individually write a one-sentence prediction: 'By the end of today, the one remaining idea I expect to finally make sense is ___.'	1. Display the original Timer Race Puzzle description (projector or written on board). Say verbatim: 'Before we do anything new today, I want you to travel back in time. Open your journal to your very first entry in this unit. Read what you wrote then. Do not change it — just read it.' Allow 3 minutes of silent reading. 2. Say: 'Now write two sentences only: what you believed then, and what you know now. You have 2 minutes. Go.' WAIT TIME: Allow full 2 minutes of silent writing. 3. Distribute or display the full DQB. Say: 'With your partner, go through every question on this board. Mark each one A — fully answered, B — partially answered, or C — still a mystery. You have 4 minutes.' WAIT TIME: 4 minutes pair work. 4. Cold-call 4 pairs (select mixed-ability pairs deliberately) to report one categorisation each. After each pair, ask the class: 'Does anyone disagree with that categorisation? Thumbs up if you agree, thumbs sideways if you are not sure.' 5. Record class consensus DQB categories on a master list on the board. Say: 'Hold onto the C items — we are going to close every single one of them today.' 6. Prompt individual prediction writing. Cold-call 3 learners to share their one-sentence prediction. Do not correct — affirm: 'That is exactly the kind of honest thinking mathematicians use.'	Strategy: 'Then vs. Now Metacognitive Audit'. By re-reading their Lesson 1 entry and explicitly comparing it to current understanding, learners activate prior knowledge, surface misconceptions they have already resolved, and identify genuine remaining gaps. This primes them for deep engagement in the final explanation phase. It also builds mathematical identity — learners see themselves as people who have genuinely grown in understanding.	Observable indicator: Every learner has written two sentences in their journal (then vs. now). Teacher circulates and scans for learners who write identical statements — this signals a learner who may not have consolidated understanding and needs targeted support during Phase 3. DQB categorisation reveals class-level gaps: any question that more than 3 pairs mark 'C' is flagged for explicit address in Phase 3.
Observe Phase 📹 **VIDEO:** Mean value	Learners receive a structured problem card (read aloud by the teacher and displayed on the board) describing the following	1. Distribute the problem card. Read it aloud slowly and clearly. Say: 'This is a real situation that plays out on Kenyan highways	Strategy: 'Problem Decomposition Mapping'. Learners break a complex multi-step problem into its component parts before attempting	Observable indicator: Each group's data table correctly converts 90 km/h → 25 m/s, 60 km/h → 16.67 m/s, 120 km/h → 33.33 m/s.

theorem application Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	scenario: A matatu travelling at 90 km/h on the Nakuru–Nairobi highway is behind a truck travelling at 60 km/h. The matatu driver wants to overtake. The oncoming lane is clear for exactly 400 metres ahead. The matatu must accelerate uniformly from 90 km/h to 120 km/h to complete the overtake. The truck is 15 metres long. The matatu is 5 metres long. Assume both vehicles travel in a straight line. Learners work in groups of 3–4 to gather and organise the given information — converting units, identifying known and unknown variables, and drawing a labelled diagram of the scenario. They do NOT yet solve; this phase is purely about reading the problem carefully, identifying the evidence (given data), and representing the scenario visually. Each group produces: (1) a unit-converted data table (all values in m/s and metres), (2) a labelled diagram showing the two vehicles and the 400 m clear zone, and (3) a list of the sub-questions they need to answer in order to determine whether the overtake is safe.	every day. Mathematics can tell us whether this overtake is safe or dangerous. Your job right now is NOT to solve — it is to understand the problem deeply.' 2. Say: 'Step one: convert every speed to metres per second. Step two: draw the scene. Step three: write down every sub-question you need to answer. You have 10 minutes.' WAIT TIME: Allow full 10 minutes of group work. Circulate and observe — do NOT give answers. Ask probing questions only: 'What does 90 km/h mean in m/s?', 'What total distance must the matatu travel to completely pass the truck?', 'What two things must you calculate to decide if the overtake is safe?' 3. After 10 minutes, cold-call 2 groups to share their data tables. Ask the class: 'Did anyone get a different conversion? Let us check together.' Correct unit conversion errors whole-class before proceeding. 4. Cold-call 1 group to share their labelled diagram on the board. Ask: 'What distance must the matatu actually travel to be completely past the truck? Think about the lengths of both vehicles.' Allow learners to debate — WAIT TIME 15 seconds before accepting answers. 5. Say: 'You now have all the ingredients. In the next phase, you will cook the meal.'	calculation. This mirrors the practice of real engineers and transport planners. By first drawing the scenario and listing sub-questions, learners avoid the common error of jumping to a formula without understanding what is being asked — a critical mathematical practice (NGSS SEP 5: Using Mathematics and Computational Thinking).	Groups that cannot perform the conversion are identified immediately and the teacher provides a brief whole-class re-teach of unit conversion (no more than 2 minutes) before proceeding. Groups' sub-question lists should include at minimum: (a) time to accelerate, (b) distance covered by matatu during overtake, (c) total distance needed to clear the truck, (d) comparison of available vs. required distance.
Explain Phase 📹 **VIDEO:** Mean value theorem application Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups now solve the Nakuru–Nairobi overtaking problem in full. They must: (1) Calculate the time for the matatu to accelerate from 25 m/s to 33.33 m/s given that a reasonable uniform acceleration is 2 m/s² — using a = (v−u)/t. (2) Calculate the total distance the matatu travels during this acceleration phase — using s = ut	1. Say: 'You have 20 minutes to solve the full problem. Every group must produce: the calculations, the velocity-time graph, and the written verdict. Every member must be able to explain any part of the solution.' WAIT TIME: 20 minutes group work. Circulate actively. For groups stuck on step (3), ask: 'The matatu must be completely past	Strategy: 'Multi-Representational Convergence'. Learners solve the same problem through three representations simultaneously — numerical calculation, algebraic formula application, and graphical construction (velocity-time graph). When all three representations yield consistent answers, learners experience the power of	Observable indicators: (1) Velocity-time graph has correctly labelled axes (velocity in m/s on y-axis, time in seconds on x-axis), a straight line with positive gradient for the acceleration phase, and the gradient is calculated and labelled as 2 m/s². (2) The area under the graph is calculated and matches the algebraic displacement value

📖 **HTML:** [Linear Equations: Deep Dish Pizza (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	+ ½at². (3) Calculate the total distance the matatu needs to travel to completely clear the truck (matatu length + truck length + safe buffer of 10 m = 30 m total clearance). (4) Compare the distance available (400 m) with the distance required, accounting for the fact that the truck is also moving forward during the overtake. (5) Construct a velocity-time graph for the matatu's motion during the overtake on graph paper, labelling axes, gradient (= acceleration), and the area under the graph (= displacement). (6) Write a final written verdict: 'Is this overtake mathematically safe?' with a justification referencing specific calculated values. After group work (20 minutes), two groups present their solution and velocity-time graph to the class. The class critiques each presentation using the sentence frames: 'We agree with ___ because ___.' and 'We want to challenge ___ because ___.' The teacher then delivers a 7-minute Final Explanation, explicitly connecting every concept from the unit back to the anchoring Timer Race Puzzle and to the driving question.	the truck — what distance does the front of the matatu need to travel before the back of the matatu clears the front of the truck?' For groups stuck on step (4), prompt: 'While the matatu is overtaking, the truck is not standing still. How far does the truck move in that same time?' 2. After 20 minutes, cold-call 2 groups to present. For each presentation: (a) Ask the presenting group to point to specific values on their velocity-time graph as they explain. (b) After presentation, ask class: 'Using our sentence frames — agree or challenge? You have 10 seconds to decide.' Cold-call 3 learners for responses. WAIT TIME: 15 seconds between challenge statements to allow the presenting group to respond. 3. Deliver the 7-minute Final Explanation. Use the following structure and say these phrases verbatim at key moments: — 'On Day 1, you were puzzled: same distance, same time, different places. Now you know why — distance is scalar, displacement is vector.' — 'The matatu's speedometer shows 90 km/h. That is speed — a scalar. But the direction is Nairobi, not Nakuru. That direction is what turns speed into velocity.' — 'When the driver accelerates to overtake, the velocity changes. The rate of that change — that is acceleration. And today you calculated it, graphed it, and used it to make a safety decision.' — 'Eliud Kipchoge runs 42 km in the Berlin Marathon. His distance is 42 km. His displacement? Almost zero — he ends near where he started. Same number, completely different	mathematical coherence. This strategy (NGSS SEP 2: Developing and Using Models; SEP 5: Using Mathematics) deepens conceptual understanding far beyond procedural fluency. The Final Explanation narrative explicitly ties representations back to the anchoring phenomenon, completing the storyline arc.	from s = ut + ½at². (3) Written verdict references specific numerical values (not vague statements). (4) During peer critique, learners use mathematical language (velocity, acceleration, displacement) rather than colloquial terms. Any group whose graph gradient does not match the stated acceleration value (2 m/s²) is flagged for immediate targeted feedback.

		mathematical meaning.' — 'The Timer Race puzzle is now solved. You have the language. You have the tools. You are mathematicians of motion.' 4. After the Final Explanation, ask: 'Is there any question remaining on our DQB that has not been answered?' Cold-call 3 learners. For each remaining question, invite a peer — not the teacher — to answer first.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Mean value theorem application Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The full DQB — all questions gathered across Lessons 1–9 plus any added today — is displayed. Learners individually spend 4 minutes writing a final answer (2–3 sentences) in their journals for the two KICD Key Inquiry Questions: (1) How do distance, displacement, speed, velocity, and acceleration differ from one another, and how are they calculated? (2) How can graphs be used to represent and interpret the motion of objects moving in a straight line? Pairs then share their answers and agree on a combined 'best answer' to contribute to a class consensus. The teacher facilitates a whole-class 'DQB closing ritual': each original question card on the board is physically moved (or marked) to the 'ANSWERED' column by a nominated learner who reads the answer aloud. Questions that remain partially answered are explicitly acknowledged and pointed toward future learning (e.g., 'This question about 2D motion will be explored in further physics studies').	1. Say: 'Four minutes — write your best answer to both key inquiry questions. This is your final chance to put into your own words what you now know. No copying from notes — use your own thinking.' WAIT TIME: 4 minutes silent individual writing. 2. Say: 'Share with your partner. Together, write the single best sentence that answers each question.' Allow 3 minutes pair discussion. 3. Facilitate DQB closing. For each major question category on the DQB, nominate a different learner to come forward, read the question, and read the class's consensus answer. Say: 'You asked this question in Week 1. Read what we have learned.' After each answer is read, ask the class: 'Thumbs up if you agree this question is now answered.' 4. For any question receiving less than 80% thumbs-up, say: 'This one still has some mystery. Let us hear from someone who can add to the answer.' Cold-call 2 learners. 5. At the end, say: 'Look at this board. Every question you asked — you answered. That is what mathematicians do: they ask hard questions and then they work until they have answers.'	Strategy: 'Public Knowledge Construction via DQB Closure'. The DQB closing ceremony transforms individual learning into publicly shared, socially validated knowledge. By physically moving question cards to the 'answered' column, learners experience the tangible progression of their understanding. The ritual nature of the ceremony reinforces the narrative arc of the unit and gives learners a sense of intellectual accomplishment. This strategy also models authentic scientific and mathematical practice: inquiry begins with questions and concludes when evidence-based answers are established.	Observable indicator: Learners' journal answers to the two key inquiry questions must include: (a) explicit comparison of at least two scalar-vector pairs (e.g., distance vs. displacement, speed vs. velocity), (b) at least one formula stated correctly, and (c) at least one reference to what a graph gradient or area represents. Journals are collected at the end of the lesson for teacher review. Any learner whose answer omits vectors entirely is identified for follow-up consolidation.

Model Building Phase 📹 **VIDEO:** Mean value theorem application Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative motion model — the annotated diagram/graph that has been revised across every lesson in the unit, beginning with a simple arrow sketch in Lesson 1 and growing to include distance-time graphs, velocity-time graphs, acceleration calculations, and vector annotations. They spend 8 minutes completing their final revision: adding any remaining annotations, ensuring all key vocabulary is labelled, and writing a 3-sentence 'Model Story' at the bottom of their model that explains: (1) What the Timer Race Puzzle was really showing us, (2) What mathematical tools we needed to fully explain it, and (3) How our model changed across the unit. Learners then display their final models (desk gallery or wall display) and do a 3-minute silent gallery walk, placing a small sticker dot on the one model (not their own) that best explains the phenomenon. The learner whose model receives the most dots shares their model story with the class. Finally, each learner independently completes a written Exit Reflection (3 minutes) in their journal: 'The most powerful mathematical idea I learned in this unit is ___ because ___. If I were explaining linear motion to a Form 1 student, I would start by telling them ___.'	1. Say: 'This is the moment. Get out your cumulative model — every version, every revision. You have 8 minutes to make your final additions and write your three-sentence Model Story at the bottom. Make it count.' WAIT TIME: 8 minutes individual work. Circulate. For learners who are unsure what to add, ask: 'Does your model show direction? Does it show what acceleration looks like on a velocity-time graph? Does it connect back to the timer race?' 2. Facilitate the gallery walk. Say: 'Walk in silence. Read every model. Place your dot on the one model — not your own — that you think best tells the story of linear motion.' Allow 3 minutes. 3. Identify the model with the most dots. Invite that learner to stand and read their Model Story aloud. After they finish, ask the class: 'What made this model powerful? Give one specific reason.' Cold-call 3 learners. 4. Say: 'Now, final reflection. Open your journal to a fresh page. Complete these two sentences — in your own words, no copying: The most powerful mathematical idea I learned in this unit is ___ because ___. If I were explaining linear motion to a Form 1 student, I would start by telling them ___. You have 3 minutes. Begin.' WAIT TIME: 3 minutes strict silent writing. 5. Collect journals. As learners pack up, say verbatim: 'Ten lessons ago, you watched three people walk across a compound and you were puzzled. Today, you can write the equations, draw the graphs, and use the mathematics to make a safety decision on a Kenyan highway. That is what growth in	Strategy: 'Cumulative Model Comparison — L1 vs. Final'. By physically holding their Lesson 1 model alongside their final model and writing the Model Story, learners make their own conceptual growth visible and explicit. Research shows that making learning visible to the learner themselves is one of the highest-impact strategies for consolidation and long-term retention. The gallery walk adds a social dimension: learners evaluate peers' models using the same conceptual criteria they have internalised, reinforcing mastery. The exit reflection (NGSS SEP 8: Obtaining, Evaluating, and Communicating Information) anchors the learning in the learner's own voice.	Observable indicators: (1) Final cumulative model includes: at least one distance-time graph AND one velocity-time graph, vector arrows with direction labels, at least three key vocabulary terms (displacement, velocity, acceleration) correctly used in annotations, and a Model Story that explicitly references the Timer Race Puzzle. (2) Exit reflection sentences are written in learner's own language (not copied from the board) and include a specific mathematical idea (not a vague statement like 'I learned about motion'). (3) The gallery walk dot distribution provides the teacher with a class-level snapshot of which models — and therefore which learners — have achieved the deepest integration. Journals are reviewed by the teacher before the next lesson to identify any learner requiring remediation before the summative assessment.

		mathematics looks like. Well done.'		

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions before the summative assessment: 1. PHENOMENON CLOSURE: Did every learner explicitly connect the Timer Race Puzzle to the concept of vectors (displacement vs. distance) by the end of the lesson? Were there learners who could solve the highway calculation but still described velocity and speed as interchangeable? These learners have procedural fluency without conceptual depth — they need targeted questioning before the assessment. 2. MODEL QUALITY: During the gallery walk, note the range of model quality. Did most learners' final models include both types of graphs and vector annotations? If the majority of models show only one graph type, consider whether earlier lessons gave sufficient time for model building or whether graph construction needs reinforcement. 3. DQB GAPS: Were there questions on the DQB that the class could not answer confidently (fewer than 80% thumbs-up)? Record these questions. They represent genuine learning gaps that should be addressed in assessment feedback or in the introduction to the next sub-strand. 4. HIGHWAY PROBLEM: Did groups correctly account for the truck's motion during the overtake (i.e., did they subtract the distance the truck travels from the 400 m available)? This is the most common error — conflating a fixed reference frame with a moving one. If most groups missed this, plan a brief whole-class follow-up discussion at the start of the next lesson or in assessment review. 5. EQUITY CHECK: Which learners did NOT participate orally across the five phases? Review cold-call records. Ensure that assessment tasks offer multiple modes of expression (written, graphical, oral) so that learners who are quieter in class have equitable opportunity to demonstrate understanding. 6. PERSONAL REFLECTION: Did the Final Explanation narrative feel natural and convincing, or did it feel rushed? The seven-minute Final Explanation is the emotional and intellectual climax of ten lessons — if it felt rushed, consider restructuring Phase 3 timing in future delivery by shortening group solving time by 3 minutes and protecting the explanation time more firmly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners returned to the Timer Race Puzzle and the full Driving Question Board (DQB), categorised accumulated evidence and questions as fully answered, partially answered, or still uncertain, and identified remaining conceptual gaps. They then received and decomposed a rich real-world problem: determining whether a matatu can safely overtake a truck on the Nakuru–Nairobi highway given specific velocity, acceleration, and distance data.
What did I learn?	Learners integrated all unit concepts — scalar vs. vector quantities (distance/displacement, speed/velocity), the formula $v = s/t$, the acceleration formula $a = (v-u)/t$, equations of uniform motion ($s = ut + ½at^2$), and the construction and interpretation of velocity-time graphs (gradient = acceleration; area under graph = displacement) — to solve a multi-step real-world safety problem. They verified that all three mathematical representations (algebraic, numerical, graphical) yield consistent results, experiencing mathematical coherence. They completed the DQB and delivered a final explanation tying every concept back to the anchoring phenomenon.
How does this explain the	The Timer Race Puzzle is now fully resolved: the same numerical value for distance, speed, or time can describe entirely different

phenomenon?	physical situations because motion in a straight line requires direction — making displacement, velocity, and acceleration vector quantities. Mathematics gives us the tools (formulae, graphs, and models) to not only describe motion precisely but to predict and evaluate it — determining whether a highway overtake is safe, understanding why Eliud Kipchoge's displacement after a marathon is nearly zero despite running 42 km, and explaining why a matatu's speedometer alone cannot tell you where in Kenya you are headed. Direction always matters.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.